

Navier–Stokes source propagation and the finite coupled bridge

Original third-growth scales, exact pulse map, and complete residual

This reader contains the complete preceding operator, pulse, covariance, source-transport and radial-moment proofs. The new parts retain the original third-growth seed and evolving parent, prove all amplitude derivative costs, identify the exact source clock and tangent frame, construct an invertible map to the actual source homogeneous pulse, and calculate the entire compact curl candidate and its force increment.

The released source theorem is attributed to the supplied manuscript; this lane has not independently certified its whole proof. Its positive swirl path and the exact negative reflected path retain their respective forces. The distinct finite coupled candidate has the explicit residual proved here. The companion 102-page reader contains its preceding original-stage proofs. No infinite modified coupled construction or uniform terminal force estimate is asserted by the finite pulse map.

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Part I

Complete preceding bridge and actual source propagation

1 Exact operator bridge to the released Navier–Stokes profiles

This bounded comparison uses the locally saved released Navier–Stokes source (the source pages are 7–15 and 24–27 for the profile, stress and pulse construction). Its stated physical viscosity is denoted by $\nu_{\text{NS}} > 0$. The Boussinesq lane variables and physical viscosity remain d, μ and $\nu = \sqrt{A_0}$; none is identified with ν_{NS} .

Use right-handed cylindrical coordinates (r, ϑ, z) and define

$$y_1 = z, \quad y_2 = \frac{r^2}{2} > 0, \quad v = (u^z, ru^r) = J \nabla_y \Psi, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The meridional divergence identity is $\partial_{y_1} v_1 + \partial_{y_2} v_2 = 0$. Define the circulation and meridional vorticity variables

$$\Gamma = ru^\vartheta, \quad \xi = \frac{\omega^\vartheta}{r}, \quad h = \frac{\Gamma}{2y_2} = \frac{u^\vartheta}{r}, \quad Q = \partial_{y_1}^2 + 2y_2 \partial_{y_2}^2, \quad M = Q + 4\partial_{y_2}.$$

The azimuthal momentum and meridional vorticity rows of the cylindrical equations (with arbitrary smooth force components) become, on $y_2 > 0$,

$$D_v \Gamma = \nu_{\text{NS}} Q \Gamma + r f^\vartheta, \quad (1)$$

$$D_v \xi = \nu_{\text{NS}} M \xi + \partial_{y_1} (h^2) + \text{curl}_y (f^z, f^r/r), \quad (2)$$

where $D_v = \partial_t + v \cdot \nabla_y$ and $\text{curl}_y(a, b) = \partial_{y_1} b - \partial_{y_2} a$.

Indeed, $\partial_r = r \partial_{y_2}$ and $\partial_r^2 + r^{-1} \partial_r = 2y_2 \partial_{y_2}^2 + 2\partial_{y_2}$. For Γ , the azimuthal viscous operator is $\partial_r^2 - r^{-1} \partial_r + \partial_z^2 = Q$. For $r\xi = \omega^\vartheta$, the cylindrical operator $(\partial_r^2 + r^{-1} \partial_r - r^{-2} + \partial_z^2)(r\xi)$, divided by r , is $M\xi$; explicitly the radial terms are $2y_2 \xi_{y_2 y_2} + 4\xi_{y_2}$. The centrifugal term is $r^{-2} \partial_z((u^\vartheta)^2) = \partial_{y_1}(\Gamma^2/(4y_2^2)) = \partial_{y_1}(h^2)$. These calculations prove every coefficient in (1)–(2), including the r force conversion.

The velocity–vorticity relation is also retained. Since $u^z = -\Psi_{y_2}$ and $u^r = \Psi_{y_1}/r$,

$$\xi = \frac{\partial_z u^r - \partial_r u^z}{r} = \frac{\Psi_{y_1 y_1}}{2y_2} + \Psi_{y_2 y_2} = \frac{Q\Psi}{2y_2}.$$

Thus the two transport rows do not by themselves replace the elliptic velocity reconstruction. Conversely, on a simply connected meridional domain, a smooth Ψ, Γ satisfying this relation and both transport rows recovers u . Define its momentum residual without pressure as $a = \partial_t u + (u \cdot \nabla)u - \nu_{\text{NS}} \Delta u - f$. The circulation row is $a^\vartheta = 0$, while the vorticity row is $\partial_z a^r - \partial_r a^z = 0$. The one-form $a^r dr + a^z dz$ is therefore closed, and $p = -\int (a^r dr + a^z dz)$ along a path from a fixed base point is path independent. Its derivatives are $p_r = -a^r$, $p_z = -a^z$. This proves the local momentum reconstruction, with pressure unique up to a function of time; boundary data are not removed.

The change from circulation to swirl rate is invertible on the exact retained domain: $\Gamma = 2y_2 h$. Product differentiation gives

$$Q(2y_2 h) = 2y_2 M h, \quad D_v(2y_2 h) = 2y_2 D_v h + 2v_2 h.$$

Therefore (1) is equivalent to

$$D_v h = \nu_{\text{NS}} M h - \frac{v_2}{y_2} h + \frac{f^\vartheta}{r}. \quad (3)$$

The source profile uses precisely this relation: in its similarity variables, E is the azimuthal velocity profile and $H_{\text{src}} = \sqrt{2X}E$ is the scaled circulation profile (source equations (4.3), (4.8)). Since $r = \sqrt{2qX}$ and $u^\vartheta = q^{-A}E$, its physical circulation is $\Gamma = q^{1/2-A}H_{\text{src}} = q^{-h_{\text{src}}}H_{\text{src}}$, where the source exponents satisfy $A = 1/2 + h_{\text{src}}$. Here $q > 0$ is the geometric similarity coordinate and $h_{\text{src}} > 0$ is an exponent; neither is a signed amplitude scale. This is the exact scale dictionary; it does not identify H_{src} with the Boussinesq scalar amplitude.

The sign of the blowout amplitude must be kept separate from that positive geometric coordinate. The axisymmetric equations have the exact involution

$$(u^r, u^\vartheta, u^z, p; f^r, f^\vartheta, f^z) \longmapsto (u^r, -u^\vartheta, u^z, p; f^r, -f^\vartheta, f^z).$$

It sends (Γ, h) to $(-\Gamma, -h)$, leaves ξ and the meridional flow unchanged, and preserves every displayed equation: the only quadratic swirl term is $\partial_{y_1}(h^2)$, while the azimuthal equation is linear in (Γ, f^ϑ) . Consequently a source-selected positive profile $E > 0$ (source pp. 7–9) does not justify assuming that every blowout branch is positive; its reflected branch has $E < 0$, $\Gamma < 0$ and the same blowup magnitude. The exponent $q^{-h_{\text{src}}}$ controls the magnitude in either branch. The source's positive covariance coefficients and viscosity are separate construction choices, not a sign claim about the velocity amplitude.

For comparison with the lane's controlled scalar equation, set $B = h^2$. The globally smooth square equation, including its geometric and viscous terms, is

$$D_v B = \nu_{\text{NS}} M B - 2\nu_{\text{NS}} |\nabla h|_Q^2 - \frac{2v_2}{y_2} B + \frac{2h}{r} f^\vartheta, \quad |\nabla h|_Q^2 = h_{y_1}^2 + 2y_2 h_{y_2}^2. \quad (4)$$

This follows by multiplying (3) by $2h$ and using $M(h^2) = 2hMh + 2|\nabla h|_Q^2$. At zeros of h equation (4) remains smooth; dividing by B is unnecessary and would lose that domain. On a component where $B > 0$, the equivalent expression is $D_v B = \nu_{\text{NS}} M B - \nu_{\text{NS}} |\nabla B|_Q^2 / (2B) - 2v_2 B / y_2 + 2h f^\vartheta / r$.

The exact relation to the Boussinesq lane is consequently an operator correspondence, not a scalar conjugacy. The lane's retained-viscosity scalar row has $D_v \Theta = d\Delta\Theta + f_\Theta$ in Cartesian material coordinates. A direct identification $\Theta = B$ would require, on the same physical domain, both $M = \Delta$ after the nonlinear coordinate pullback and $-2v_2 B / y_2 - 2\nu_{\text{NS}} |\nabla h|_Q^2 + 2h f^\vartheta / r$ to be absorbed into one prescribed force. The first condition fails as an operator identity: $M = \partial_{y_1}^2 + 2y_2 \partial_{y_2}^2 + 4\partial_{y_2}$ has variable metric and drift, whereas the lane's Δ is the Euclidean Cartesian Laplacian. The geometric, gradient and force terms must therefore be retained or calculated in the particular source state. Nonconstant profiles alone would not prove that such terms never cancel. The exact bridge is (1)–(4) with the full residual terms retained. It neither claims that the lane's controls are the source pulses nor changes the source's pressure reconstruction.

For the meridional velocity, the source's pulse construction supplies the following further exact local dictionary (source pp. 11–15 and 74–77). If w is a divergence-free pulse and π its pressure increment, then

$$R(u_B + w, p_B + \pi) = R(u_B, p_B) + L_{u_B}(w, \pi) + \nabla \cdot (w \otimes w),$$

$$L_{u_B}(w, \pi) = \partial_t w + (u_B \cdot \nabla) w + (w \cdot \nabla) u_B - \nu_{\text{NS}} \Delta w + \nabla \pi, \quad \nabla \cdot w = 0.$$

For the moving local phase with wavevector n and transverse amplitude t_m , the source equation is

$$t'_m + \mathcal{K} t_m + m^2 d_N t_m + i k m n \pi_m = -f_m, \quad n \cdot t_m = 0, \quad d_N = \varepsilon k^2 |n|^2,$$

with the source's moving shear matrix $\mathcal{K} = \begin{pmatrix} 0 & -2F & 0 \\ 2F + RF_R & 0 & 0 \\ G_R & 0 & 0 \end{pmatrix}$. The transverse projection uses

$$A_\Phi = -\mathcal{K} + \frac{n(n^T \mathcal{K} - n'^T)}{|n|^2}.$$

These are exact source pulse equations, including viscosity and pressure projection. Their quadratic covariance is chosen to cancel the annular stress; the source explicitly retains curl, cutoff, and correction interactions. The Boussinesq lane's two-dimensional pair and its full profile residuals have an exact algebraic form, but no identification with this three-dimensional pulse equation follows without an additional embedding that supplies the source's cylindrical geometry, moving plane, covariance cone and correction cycle. That embedding is not asserted here.

The current bridge therefore proves a precise morphism of variables, operators, forces and pulse residuals and records the exact obstruction to a direct scalar identification. It establishes the next programme datum while preserving the unproved status of any singular limit or third return.

2 Scope and source equations

This note records one local, exact comparison. It does not identify the coupled Boussinesq stages with the released Navier–Stokes construction and makes no claim about a global Navier–Stokes solution. The source is the released PDF `openai_navier_stokes.pdf`, SHA-256 `8c8a94ad9ac824c8b605b9827cadf7beaca48bd10b380de3cfc872a2c37afa81`. The source statements used here are Section 7, equations (7.2)–(7.8), (7.12)–(7.22), and Section 9, equations (9.1)–(9.2), on PDF pages 74–75, 77–81, and 101–103. In the source notation, on one lifted pulse rectangle,

$$\Phi = p\theta + p_z Z/\varepsilon + x_0 R - v(pF + p_z G), \quad n = \nabla_* \Phi, \quad (5)$$

with

$$n = (x_0 - v(pF_R + p_z G_R), p/R, p_z - \varepsilon v(pF_Z + p_z G_Z)). \quad (6)$$

For a nonzero integer harmonic m , the exact principal pulse equations are

$$n \cdot t_m = 0, \quad t'_m + K t_m + m^2 d_N t_m + i k m n \pi_m = -f_m, \quad d_N = \varepsilon k^2 |n|^2, \quad (7)$$

where a prime is the pulse-coordinate derivative and

$$K = \begin{pmatrix} 0 & -2F & 0 \\ 2F + RF_R & 0 & 0 \\ G_R & 0 & 0 \end{pmatrix}, \quad A_\Phi = -K + \frac{n(n^T K - (n')^T)}{|n|^2}. \quad (8)$$

The source pressure coefficient is exactly

$$\pi_m = \frac{i}{k m |n|^2} (n \cdot K t_m - (n') \cdot t_m + n \cdot f_m). \quad (9)$$

For $m = 1$, a real homogeneous pulse t_1 has the source energy identity (7.22):

$$\frac{1}{2}(|t_1|^2)' = -g \cdot (t_{1,r}(t_{1,\theta}, t_{1,z})) - \varepsilon k^2 |n|^2 |t_1|^2, \quad (10)$$

where $g = (RF_R, G_R)$ in the tangential coordinates. Thus the source pulse retains both shear transfer and viscous dissipation.

3 The coupled physical pair

For one activated layer of the coupled calculation, put

$$a_c = -\frac{J\zeta \cdot G_{<q}}{Kr^2}, \quad b_c = K\zeta_1, \quad \lambda_c^\Theta = \kappa K^2 r^2, \quad \lambda_c^\Omega = \varepsilon_c K^2 r^2. \quad (11)$$

Its exact retained-viscosity pair, including optional scalar and vorticity controls, is

$$\dot{y} = C_c y + c_c, \quad y = \begin{pmatrix} \Theta \\ \Omega \end{pmatrix}, \quad C_c = \begin{pmatrix} -\lambda_c^\Theta & a_c \\ b_c & -\lambda_c^\Omega \end{pmatrix}, \quad c_c = \begin{pmatrix} c_\theta \\ c_\omega \end{pmatrix}. \quad (12)$$

Here $G_{<q}$, ζ , and r evolve with the inherited background and phase ODEs; in particular no fixed-background substitution is made. Let v be a source pulse coordinate on an interval and let $q_v = dt/dv$ describe its physical-time parametrization. The forward source parametrization has $q_v > 0$; the algebra below also retains a negative orientation without changing the amplitude sign. Define $y'(v) = q_v(C_c y + c_c)$. Here K, r are the coupled frequency and phase length, not the source carrier k and cylindrical radius R . The original A_0, λ_0, μ and time scale $\nu = \sqrt{A_0}$ are unchanged. The source chart parameter ε is not identified with either physical diffusivity.

4 Exact local residual map

Let $B(v) \in \mathbb{R}^{3 \times 2}$ be a C^1 tangent frame with full column rank and $n(v)^T B(v) = 0$, so its range is exactly $n(v)^\perp$. Let $B_\ell(v)$ be any left inverse, $B_\ell B = I_2$; then BB_ℓ acts as the identity on $n^\perp$. Let $L(v) \in \mathbb{R}^{2 \times 2}$ be any C^1 coefficient map and embed the coupled pair into a transverse source amplitude by

$$t(v) = B(v)L(v)y(v). \quad (13)$$

For an embedding with a non-tangent frame the transversality condition would be

$$n(v)^T B(v)L(v)y(v) = 0. \quad (14)$$

For the tangent frame used here, the condition holds for every L, y . Set

$$H(v) = B_\ell(v)(A_\Phi(v)B(v) - B'(v)). \quad (15)$$

Then the exact projected source residual of the embedded coupled pair is

$$\mathcal{E}_{L,c} := B_\ell \text{proj}_{n^\perp} [t' - A_\Phi t + m^2 d_N t + f_m] = [L' + q_v L C_c - H L + m^2 d_N L] y + q_v L c_c \quad (16)$$

$$+ B_\ell \text{proj}_{n^\perp} f_m. \quad (17)$$

In particular, with $L = I$ and no added source, the defect is the explicit matrix expression

$$\mathcal{E}_{I,0} = (q_v C_c - H + m^2 d_N I_2) y. \quad (18)$$

This gives a local residual relation with every actual coefficient retained.

Proof. Differentiate (13): $t' = B'Ly + BL'y + BLy'$. Since $y' = q_v(C_c y + c_c)$, differentiate $n^T B = 0$ to obtain $n^T B' = -(n')^T B$. Also $n^T A_\Phi B = -(n')^T B$, directly from (8). Thus $A_\Phi B - B'$ has transverse range and equals BH . Applying B_ℓ to the transverse terms and $B_\ell \text{proj}_{n^\perp}$ to the force proves (17). More explicitly,

$$n^T(t' - A_\Phi t + m^2 d_N t) = 0, \quad n^T(t' - A_\Phi t + m^2 d_N t + f_m) = n^T f_m. \quad (19)$$

The unprojected force need not be transverse. The pressure below removes precisely this remaining normal component. $\square$

5 Pressure reconstruction and full three-dimensional residual

For any embedded transverse t and arbitrary f_m , define

$$\pi_{L,c} = \frac{i}{km|n|^2} (n \cdot Kt - (n') \cdot t + n \cdot f_m). \quad (20)$$

Then the full source residual satisfies the exact decomposition

$$\mathcal{R}_{L,c} := t' + Kt + m^2 d_N t + ikmn\pi_{L,c} + f_m \quad (21)$$

$$= \text{proj}_{n^\perp} (t' + Kt + m^2 d_N t + f_m) = B \mathcal{E}_{L,c}, \quad (22)$$

where the last equality uses the definition of H and the transverse condition. Thus the pressure removes precisely the normal component; it does not remove $\mathcal{E}_{L,c}$, the moving-frame, shear, diffusion, or coupled-background mismatch. If one elects to force the source pulse to obey the released equation, then the unique additional transverse forcing is

$$f_{m,\text{corr}} = -B \mathcal{E}_{L,c}, \quad (23)$$

added to the old f_m . The old $\mathcal{E}_{L,c}$ is evaluated before this addition, so the new projected residual is zero. Since $n^T f_{m,\text{corr}} = 0$, pressure is unchanged. This is an exact local forced relation, not an identification of the physical constructions.

6 Diffusion and energy comparison

The source diffusion multiplier is $m^2 d_N = m^2 \varepsilon k^2 |n|^2$ in (7). The coupled pair has the two retained physical losses λ_c^Θ and λ_c^Ω , which are generally unequal to $m^2 d_N$. Their signed discrepancy in (18) is explicitly

$$(m^2 d_N - q_v \lambda_c^\Theta) \Theta, \quad (m^2 d_N - q_v \lambda_c^\Omega) \Omega \quad (24)$$

when $H = 0$ and C_c is expanded entrywise; with general H the exact diagonal entries are those of $q_v C_c - H + m^2 d_N I_2$. No damping is dropped. If one compares nonnegative loss magnitudes instead, the corresponding cost gaps are $|m^2 d_N - q_v \lambda_c^\Theta|$ and $|m^2 d_N - q_v \lambda_c^\Omega|$. The required finite-interval coordinate map can be constructed explicitly. Let $I = [v_0, v_1]$, and let X_c, X_s be the identity-initial-value fundamental matrices for $D_c = q_v C_c$ and $D_s = H - m^2 d_N I_2$, respectively. For either continuous matrix D , define

$$X_D(v) = I_2 + \sum_{j \geq 1} \int_{v_0 \leq w_j \leq \dots \leq w_1 \leq v} D(w_1) \cdots D(w_j) dw_j \cdots dw_1.$$

With $M_D = \sup_I \|D\|$ and $T_I = v_1 - v_0$, the j th integral has norm at most $(M_D T_I)^j / j!$. Uniform convergence proves the integral equation $X_D = I_2 + \int_{v_0}^v D X_D$ and hence $X_D' = D X_D$. The same iteration for $Z' = -ZD$, $Z(v_0) = I_2$ gives $Z X_D = I_2$ by differentiation, proving invertibility on I . For a specified invertible L_0 , set

$$L = X_s L_0 X_c^{-1}, \quad L^{-1} = X_c L_0^{-1} X_s^{-1}.$$

This has norm at most $\|L_0\| e^{(M_c + M_s) T_I}$; its inverse has the same bound with L_0^{-1} . Differentiation proves

$$L' = H L - q_v L C_c - m^2 d_N L, \quad (25)$$

so the homogeneous embedded pair has zero projected residual. This proves an actual finite-interval conjugacy along the retained coefficient path. It does not make the source phase, pressure, cutoff, or nonlinear covariance equal to the coupled Boussinesq fields. For general controls c_c , the forced term is $q_v L c_c$ in (17).

For an arbitrary embedded field (possibly complex), the exact energy balance retains its full defect:

$$\begin{aligned} \frac{1}{2}(|t|^2)' = & -\operatorname{Re}(\bar{t}^T \mathbf{K} t) - m^2 d_N |t|^2 - \operatorname{Re}(\bar{t}^T f_m) \\ & + \operatorname{Re}(\bar{t}^T B \mathcal{E}_{L,c}). \end{aligned} \quad (26)$$

This follows by taking the real Hermitian product of (22) with t . Its pressure term vanishes because n is real and $n^T t = 0$. The $2F$ part of $\mathbf{K}$ is skew-symmetric, hence its real work is zero; the shear work equals $\operatorname{Re}(\bar{t}_r(g_\theta t_\theta + g_z t_z))$. Only for a zero residual, zero force, $m = 1$, and real t does this reduce to (10). The real coupled pair has

$$\frac{1}{2} \frac{d}{dt} |y|^2 = -\lambda_c^\Theta \Theta^2 - \lambda_c^\Omega \Omega^2 + (a_c + b_c) \Theta \Omega + c_\theta \Theta + c_\omega \Omega, \quad (27)$$

which is a two-component physical energy identity. Equations (17) and (27) exhibit exactly where the source's three-dimensional shear flux and the coupled pair's two-dimensional exchange differ.

7 Proven scope

The source uses a cylindrical, localized, divergence-free curl construction (Section 7.4 and Section 9, PDF pages 85–87 and 102–103); the displayed principal pulse equation is only one component of that construction. The comparison above proves an exact local map and its full pressure-projected residual for every finite interval on which the denominators and frame left inverse exist. It preserves source viscosity, moving phase normal, pressure, rotation/shear matrix, and all coupled physical diffusion coefficients. It does not prove that the coupled third-growth or steering stages produce the source's pulse phase Φ , covariance, cutoff tails, or smooth terminal force. Those remain separate bridge problems.

The sign conventions in this comparison are also explicit. The quantities $q_v > 0$, $\varepsilon > 0$, and $d_N = \varepsilon k^2 |n|^2 > 0$ record the chosen orientation of physical time and the positive viscous coefficient; they are not assumptions on the sign of a pulse or blowout amplitude. The equations (7)–(9) are linear in (t_m, f_m) , so the simultaneous replacement $(t_m, f_m, \pi_m) \mapsto (-t_m, -f_m, -\pi_m)$ is an exact reflected branch. Likewise, the local map (13) is sign-equivariant under $y \mapsto -y$ and $c_c \mapsto -c_c$, with the coefficients fixed at the actual path. This is a statement about the retained linear equations, not a sign symmetry of the entire coupled nonlinear system. The source covariance is quadratic, so reversing a wave leaves that covariance unchanged. Reversing the full Navier–Stokes velocity need not reverse its force, since $(u \cdot \nabla)u$ is even in u . The full-flow map that reverses axisymmetric swirl is a spatial reflection; its complete formula is proved in the accompanying covariance bridge. No sign of a blowout or endpoint is inferred from positive geometric scales.

8 Source objects and the finite calculation

This calculation uses the preserved 165-page supplied manuscript *Finite Time Blowup for Navier–Stokes*, source equations (7.23)–(7.42), (9.12)–(9.14). Its PDF SHA-256 is

The source pulse fields, auxiliary rectangles, phases, and leading covariance columns are the given objects; their global construction is source input. Every finite map asserted below is proved here. The existing coupled-stage proofs remain in the complete 102-page reader. The companion pulse note proves the exact frame residual and a constructed finite-interval intertwiner along its evolving coefficients. The source uses positive $q(z, t)$, band scales $Q_\beta = 2^{-\ell_\beta}$, $\varepsilon_\beta = Q_\beta^h$, $A = 1/2 + h$; none is set to one here or identified with the original coupled $A_0, \lambda_0, \mu, \nu = \sqrt{A_0}$ or its physical diffusivities.

All operations in this note are on a compact set inside $q > 0$ and the source active annulus. There are finitely many relevant bands and locally finite source labels. Supports of newly prescribed inputs are strictly inside the annulus, so no lower bound on a leading amplitude at a vanishing shell edge is assumed. Endpoint and shell-edge estimates are not inferred from compact-interior estimates.

9 Physical curl of the actual retained amplitude

Let $y = (\Theta, \Omega)^T$ be the actual smooth retained coupled solution, $y' = q_v(C_c y + c_c)$, with the coefficients of the companion pulse note. Use the smooth actual coefficients on the chosen slow/auxiliary patch; all harmonic coefficients and cutoffs are independent of θ in cylindrical components, and each carrier has nonzero integer angular frequency. For its source tangent frame B , smooth coefficient map L , and a fixed smooth real compact cutoff χ of this type, put

$$t_m = \chi B L y, \quad \mathcal{E}_\chi = \chi \{ [L' + q_v L C_c - H L + m^2 d_N L] y + q_v L c_c \} + \chi' L y, \quad H = B_\ell (A_\Phi B - B').$$

Here prime differentiates along the pulse coordinate; the other slow and auxiliary derivatives remain in the spatial formulas below. Smooth parameter dependence of the companion's constructed L follows directly from its integral equations on a fixed interval: for a nonzero parameter multi-index I ,

$$\partial^I X(v) = \int_{v_0}^v \left[D \partial^I X + \sum_{0 < J \leq I} \binom{I}{J} (\partial^J D) \partial^{I-J} X \right] dw.$$

The initial parameter derivatives vanish for identity initial data. Inductively the known forcing is smooth and bounded on the compact patch; the same convergent ordered-integral iteration solves this equation. This proves every finite mixed derivative of X, X^{-1}, L . Since $n^T B = 0$, $n^T t_m = 0$. Differentiation proves $t'_m - A_\Phi t_m + m^2 d_N t_m = B \mathcal{E}_\chi$. Thus even the pulse-cutoff term is explicit.

A physical version of the exact curl construction makes the intermediate map to momentum residual unambiguous. For each chosen harmonic let $\phi(x, t) = k_\beta m \Phi_\beta(x, t, Y(x, t))$ be its real evaluated phase, let $\xi = \nabla_x \phi \neq 0$, and let $a(x, t) = Q_\beta^{-A} t_m$ be the physical transverse amplitude. The source coordinate change is $r = \sqrt{Q_\beta} R$, $z = Q_\beta^D Z$, $1 - t = Q_\beta T$, with $D = 1/2 - h$ and normalized axial derivative $D_z = \varepsilon_\beta \partial_Z$. It gives $\xi = (k_\beta m / \sqrt{Q_\beta}) n_\Phi$; all auxiliary chain derivatives are included in n_Φ . Hence $\xi \cdot a = 0$. Define in Cartesian components

$$C = \frac{i \xi \times a}{|\xi|^2}, \quad \mathcal{A} = C e^{i\phi}, \quad \nabla \times \mathcal{A} = (a + r_a) e^{i\phi}, \quad r_a = \nabla \times C. \quad (28)$$

The vector triple-product identity gives $i \xi \times C = a$, proving the last equality. It also shows that the physical potential has exactly the source factor $Q_\beta^{1/2-A}$ relative to its chart coefficient $i n_\Phi \times t_m / (k_\beta m |n_\Phi|^2)$. Cartesian differentiation of C includes derivatives of the cylindrical basis;

the same terms are the R^{-1} terms of source (7.37)–(7.38). Take real parts and sum the finite retained harmonics. This constructs a smooth compact velocity w_y with $\nabla \cdot w_y = 0$.

Retain a finite smooth source state (u_*, p_*) , including its background, actual leading waves and their curl corrections, and the real physical pressure increments obtained from the companion pulse pressure formula with $f_m = 0$:

$$\pi_{\chi, m} = \frac{i}{k_\beta m |n|^2} (n^T \mathbf{K} t_m - (n')^T t_m), \quad \pi_y = \sum_{\beta, m} \operatorname{Re}(Q_\beta^{-2A} \pi_{\chi, m} e^{i\phi_{\beta, m}}).$$

These are smooth and compact wherever t_m is. This definition fixes the pressure map without assuming the pulse residual vanishes. Define the actual candidate residual

$$\begin{aligned} R_y &= R_{\nu_{\text{NS}}}(u_* + w_y, p_* + \pi_y), \\ R_{\nu_{\text{NS}}}(u, p) &:= \partial_t u + (u \cdot \nabla)u - \nu_{\text{NS}} \Delta u + \nabla p. \end{aligned} \quad (29)$$

This is a definition by explicit differentiated fields, not an assumed source term. Its exact expansion is

$$R_y = R_{\nu_{\text{NS}}}(u_*, p_*) + \partial_t w_y + (u_* \cdot \nabla)w_y + (w_y \cdot \nabla)u_* - \nu_{\text{NS}} \Delta w_y + \nabla \pi_y + (w_y \cdot \nabla)w_y.$$

Every cutoff, parent, phase, and curl derivative occurs in this formula. Physical ν_{NS} is retained and is distinct from every source time-scaling convention. Define $\Delta R_y = R_y - R_{\nu_{\text{NS}}}(u_*, p_*)$. We correct the new compact increment; this difference has compact support since every added term does. The baseline residual remains explicitly.

The averaging needed next is the source extended-field operation, not a spatial average of the evaluated field. Evaluate derivatives on the extended coefficients first by the complete chain rule. In a fixed chart set $\mathcal{D}_i = \partial_{x_i} + \sum_\alpha (\partial_{x_i} Y_\alpha) \partial_{Y_\alpha}$ and similarly $\mathcal{D}_t$. These are the pullbacks of commuting physical derivatives, so $[\mathcal{D}_i, \mathcal{D}_j] = 0$ on their lifted domain. For smooth periodic coefficients their auxiliary derivative terms have zero Haar average. In cylindrical components angular differentiation also differentiates the basis. This defines an extended residual $\tilde{R}_y$ whose evaluation is exactly (29). Average the tangential components of the increment at fixed slow variables:

$$F_2 = \langle \Delta \tilde{R}_{y, \theta} \rangle, \quad F_1 = \langle \Delta \tilde{R}_{y, z} \rangle, \quad \langle g \rangle = \int_{\mathbb{T}^2} \frac{1}{2\pi} \int_0^{2\pi} g \, d\theta \, dY. \quad (30)$$

These F_e are smooth and compact in the chosen physical annulus. Equations (28)–(30) are the explicit intermediate map from the retained coupled pair to a signed momentum residual. They do not identify it with the original two-dimensional Boussinesq force.

10 Radial moments, support, and the original chart scales

For each fixed (z, t) , choose $0 < r_- < r_+$ containing the compact supports of both F_e and the chosen bumps in the active annulus. The source bump is

$$b_e(r, z, t) = q^{-(e+1)/2} \hat{b}_e(r/\sqrt{q}), \quad \int_0^\infty x^e \hat{b}_e(x) \, dx = 1, \quad e = 1, 2,$$

with interior support. The change $r = \sqrt{q} x$ proves $\int r^e b_e \, dr = 1$ exactly. Define

$$M_e = \int_0^\infty r^e F_e(r) \, dr, \quad \sigma_e(r) = -r^{-e} \int_0^r s^e (F_e(s) - b_e(s) M_e) \, ds. \quad (31)$$

Then

$$(\partial_r + e/r)\sigma_e = -F_e + b_e M_e, \quad \int_0^\infty r^e (F_e - b_e M_e) dr = 0. \quad (32)$$

Differentiate $r^e \sigma_e$ to prove the first identity; integrate and use the bump normalization for the second. The integral is zero below r_- and above r_+ , so σ_e has compact smooth support in the same enclosing interval. A homogeneous solution is cr^{-e} , and a compact homogeneous solution has $c = 0$; thus the primitive is unique among compact solutions of its displayed equation. Moreover a compact solution of $(\partial_r + e/r)\sigma = -F_e$ exists if and only if $M_e = 0$: necessity follows by integrating the weighted derivative, and sufficiency follows from (31). The two moments are therefore the exact surviving finite-dimensional terms.

For $C_e = \int_{r_-}^{r_+} s^e ds$, $|M_e| \leq C_e \|F_e\|_\infty$ and

$$\|\sigma_e\|_\infty \leq r_-^{-e} C_e (\|F_e\|_\infty + \|b_e\|_\infty |M_e|).$$

If $G_e = F_e - b_e M_e$, successive radial derivatives obey the exact recurrence

$$\partial_r^{j+1} \sigma_e = -\partial_r^j G_e - e \sum_{i=0}^j \binom{j}{i} (-1)^i i! r^{-i-1} \partial_r^{j-i} \sigma_e.$$

This proves every finite radial derivative bound inductively. Parameter derivatives follow by differentiating the compact integral; all derivatives of q, b_e and the moving physical data are retained. On the present compact set q is bounded away from zero. No terminal uniform estimate is implied.

The source (9.12)–(9.14) uses physical $F_e = Q^{-2A-1/2} E_e^*$, where $E_2^* = \langle E_\theta \rangle_Y$ and $E_1^* = \langle E_z \rangle_Y$. For our fields define $E_e^* = Q^{2A+1/2} F_e$ in that same convention. With $r = \sqrt{Q}R$ and Q fixed in a chart, the exact conversions are

$$\begin{aligned} M_e &= Q^{e/2-2A} M_e^*, \quad M_e^* = \int R^e E_e^* dR, \quad b_e^*(R) = Q^{(e+1)/2} b_e(\sqrt{Q}R), \\ \Sigma_e(R) &= Q^{2A} \sigma_e(\sqrt{Q}R), \quad (\partial_R + e/R) \Sigma_e = -E_e^* + b_e^* M_e^*. \end{aligned} \quad (33)$$

Substitute $r = \sqrt{Q}R$ including its measure in the moment, and differentiate Σ_e to prove each factor. The input F_e is already auxiliary independent. Taking only an averaged moment of an unaveraged $F(R, Y)$ would not suffice: $F = b_e(R) \cos(2\pi Y_1)$ has zero averaged moment but a nonzero exterior primitive at each generic Y .

11 The source covariance derivative and its signed right inverse

At each slow point let $w_{\text{tan}} = (w_\theta, w_z)$ and define

$$\mathcal{C}(w) = \langle w_r w_{\text{tan}} \rangle, \quad \mathcal{B}(w, v) = \langle w_r v_{\text{tan}} + v_r w_{\text{tan}} \rangle.$$

Direct expansion proves $\mathcal{C}(w + v) = \mathcal{C}(w) + \mathcal{B}(w, v) + \mathcal{C}(v)$ and $D\mathcal{C}(w)[v] = \mathcal{B}(w, v)$.

Use the exact source real pulses $b_\sigma = \chi_g(\xi_g) \psi(v) t_\sigma^h \cos(k\Phi_\sigma)$, $\sigma \in \{+, -\}$. They exclude the slow cutoff η_β and scalar amplitudes; χ_g is an auxiliary transverse cutoff. The two signs have disjoint auxiliary supports. Put $H_{\text{cov}} = [\mathcal{C}(b_+) \mid \mathcal{C}(b_-)]$. For slow scalar coefficients independent of both averages,

$$\mathcal{C}(a_+ b_+ + a_- b_-) = H_{\text{cov}} (a_+^2, a_-^2)^T.$$

All cross products vanish by disjointness. The factor $1/2$ from $\langle \cos^2(k\Phi_\sigma) \rangle_\theta$ is already inside the columns.

On the source interior cone the fixed primary amplitudes satisfy $a_\sigma = \sqrt{(H_{\text{cov}}^{-1}T_{0,*})_\sigma} > 0$, $T_{0,*} = (Q/q)^{A+1/2}T_0$, and $W_0 = \sqrt{\varepsilon} \sum_\sigma a_\sigma b_\sigma$. For any signed slow real vector Σ , define

$$d_\Sigma = H_{\text{cov}}^{-1}\Sigma/\varepsilon, \quad \delta a_\sigma = (d_\Sigma)_\sigma/(2a_\sigma), \quad L_\Sigma = \sqrt{\varepsilon} \sum_\sigma \delta a_\sigma b_\sigma. \quad (34)$$

It follows exactly that

$$\mathcal{B}(W_0, L_\Sigma) = \varepsilon H_{\text{cov}}(2a \odot \delta a) = \Sigma, \quad (35)$$

$$\mathcal{C}(W_0 + L_\Sigma) = \varepsilon T_{0,*} + \Sigma + \varepsilon H_{\text{cov}}(\delta a_+^2, \delta a_-^2)^T. \quad (36)$$

This proves a linear right inverse of the derivative. It is not an exact inverse of the quadratic covariance map. There is no sign or size restriction on Σ in these identities; no square root of $H_{\text{cov}}^{-1}(T_{0,*} + \Sigma/\varepsilon)$ is taken.

With Euclidean and induced operator norms and $a_{\min} = \min_\sigma |a_\sigma| > 0$, they give the explicit bound

$$\|\mathcal{C}(L_\Sigma)\| \leq \frac{\|H_{\text{cov}}\| \|H_{\text{cov}}^{-1}\|^2}{4\varepsilon a_{\min}^2} \|\Sigma\|^2. \quad (37)$$

Indeed $\|\delta a\| \leq \|H_{\text{cov}}^{-1}\| \|\Sigma\|/(2\varepsilon a_{\min})$ and $\|(\delta a_\sigma^2)_\sigma\|_2 \leq \|\delta a\|_2^2$. All higher derivative costs are explicit from Leibniz and

$$\partial^I H_{\text{cov}}^{-1} = -H_{\text{cov}}^{-1} \sum_{0 < J \leq I} \binom{I}{J} (\partial^J H_{\text{cov}}) \partial^{I-J} H_{\text{cov}}^{-1},$$

and, for any nonzero scalar a , $\partial^I(a^{-1}) = -a^{-1} \sum_{0 < J \leq I} \binom{I}{J} (\partial^J a) \partial^{I-J}(a^{-1})$. Together with (34) these recurrences give finite bounds for each order without assuming constant amplitudes or constant columns.

12 Assembly, the actual covariance change, and residual cancellation

For the physical target $\sigma = (\sigma_2, \sigma_1)$ from (31), use the source squared slow partition $\sum_\beta \eta_\beta^2 = 1$ on its support. Set

$$\begin{aligned} W_{0,\text{phys}} &= \sum_\beta Q_\beta^{-A} \eta_\beta W_{0,\beta}, \\ v_{\text{pr}} &= \sum_\beta Q_\beta^{-A} \eta_\beta L_\beta(Q_\beta^{2A} \sigma). \end{aligned} \quad (38)$$

Overlapping slow labels have disjoint auxiliary supports. Taking cross covariance and using (35) gives

$$\mathcal{B}(W_{0,\text{phys}}, v_{\text{pr}}) = \sum_\beta Q_\beta^{-2A} \eta_\beta^2 Q_\beta^{2A} \sigma = \sigma.$$

Every band scale remains distinct. Likewise the leading physical factor is $Q^{-2A} \varepsilon (Q/q)^{A+1/2} = Q^{-A+h+1/2} q^{-A-1/2} = q^{-A-1/2}$ because $A = 1/2 + h$.

Apply the exact curl (28) to each coefficient of v_{pr} , including the slow cutoffs before curling. Write the resulting divergence-free increment as $v_s = v_{\text{pr}} + r_s$. Let $t_{s,m}$ denote the chart transverse harmonic coefficients of $\eta_\beta L_\beta(Q_\beta^{2A} \sigma)$, before multiplication by Q_β^{-A} . For each such coefficient

choose $\pi_{s,m} = i(n^T K t_{s,m} - (n')^T t_{s,m}) / (k_\beta m |n|^2)$ and sum $\text{Re}(Q_\beta^{-2A} \pi_{s,m} e^{i\phi_{\beta,m}})$ to define the smooth compact π_s . Its full gradients remain below. Let the actual old wave w consist of the wave part of u_* plus the candidate w_y , including all existing curl corrections, and write $w = W_{0,\text{phys}} + e_w$. Define tensor covariance $\mathcal{T}(w) = \langle w \otimes w \rangle$ and $\mathcal{B}_T(u, v) = \langle u \otimes v + v \otimes u \rangle$. Expansion, with $e_w = w - W_{0,\text{phys}}$, proves

$$\Delta \mathcal{T} = \mathcal{B}_T(W_{0,\text{phys}}, v_{\text{pr}}) + \mathcal{B}_T(e_w, v_{\text{pr}}) + \mathcal{B}_T(w, r_s) + \mathcal{T}(v_s). \quad (39)$$

Its first radial–tangential pair is σ . All other entries and the remaining three tensors are retained. For example their radial–tangential pair has bound

$$2\|e_w\|_{L_{\theta,Y}^2} \|v_{\text{pr}}\|_{L_{\theta,Y}^2} + 2\|w\|_{L_{\theta,Y}^2} \|r_s\|_{L_{\theta,Y}^2} + \|v_s\|_{L_{\theta,Y}^2}^2,$$

by Cauchy–Schwarz. The compact-set estimates above and the explicit curl differentiate to bound all these terms at each finite order.

For clarity the complete residual after adding v_s and its selected pressure π_s is, without any averaging,

$$\begin{aligned} R_{\nu_{\text{NS}}}(u + v_s, p + \pi_s) &= R_{\nu_{\text{NS}}}(u, p) + \partial_t v_s + (u \cdot \nabla) v_s + (v_s \cdot \nabla) u \\ &\quad - \nu_{\text{NS}} \Delta v_s + \nabla \pi_s + (v_s \cdot \nabla) v_s. \end{aligned} \quad (40)$$

This follows by expanding the product and retains viscosity and every pressure term. For a symmetric tensor S the averaged tangential divergence in physical cylindrical coordinates is

$$(\langle \nabla \cdot S \rangle)_\theta = (\partial_r + 2/r) \langle S_{r\theta} \rangle + \partial_z \langle S_{z\theta} \rangle, \quad (\langle \nabla \cdot S \rangle)_z = (\partial_r + 1/r) \langle S_{rz} \rangle + \partial_z \langle S_{zz} \rangle.$$

To verify, write the Cartesian divergence in the orthonormal cylindrical basis, using $\partial_\theta e_r = e_\theta$, $\partial_\theta e_\theta = -e_r$; periodic angular derivatives average to zero and the two basis terms give $2/r$ and $1/r$ respectively. The auxiliary chain terms average to zero by periodicity, but the physical slow derivatives remain.

Let $U = u_* + w_y$, $P_y = p_* + \pi_y$, and set

$$\mathcal{L} = \partial_t v_s + (U \cdot \nabla) v_s + (v_s \cdot \nabla) U - \nu_{\text{NS}} \Delta v_s + \nabla \pi_s.$$

For a fully explicit averaged remainder, write $U = u_{\text{mean}} + w$, where u_{mean} is its angular mean, and put $\mathcal{V} = \mathcal{L} - (w \cdot \nabla) v_s - (v_s \cdot \nabla) w$. Then the average of (40) is $\langle R_y \rangle + \langle \mathcal{V} \rangle + \nabla \cdot \Delta \mathcal{T}$. Let $S_0 = \mathcal{B}_T(W_{0,\text{phys}}, v_{\text{pr}})$ and $S_{\text{rem}} = \mathcal{B}_T(e_w, v_{\text{pr}}) + \mathcal{B}_T(w, r_s) + \mathcal{T}(v_s)$. Consequently the radial divergence of the first pair in (39) changes the prescribed averaged residual F_e into $b_e M_e$ by (32). The axial tensor divergences, the other three tensors, the exact linear terms and the pressure terms of (40) remain in the new residual. Equivalently the new averaged tangential residual is exactly

$$\begin{aligned} \langle R_{\text{new}} \rangle_{\text{tan}} &= \langle R_{\nu_{\text{NS}}}(u_*, p_*) \rangle_{\text{tan}} + (b_2 M_2, b_1 M_1) + \langle \mathcal{V} \rangle_{\text{tan}} \\ &\quad + \partial_z ((S_0)_{z\theta}, (S_0)_{zz}) + (\nabla \cdot S_{\text{rem}})_{\text{tan}}. \end{aligned}$$

Every term here is defined by the displayed differentiated fields and tensors. None is assumed to vanish. The map from y through (30), (31), (34), and (38) is now an actual composite construction. It retains the signed residual produced by the coupled candidate and the source’s original physical and chart units.

13 Signs and the exact full-flow reflection

The maps $F_e \mapsto M_e \mapsto \sigma_e \mapsto L_\Sigma$ are linear and reverse sign with their inputs; $\mathcal{C}(L_\Sigma)$ is even. The two primary amplitudes may also be replaced independently by $s_\sigma a_\sigma$, $s_\sigma \in \{-1, 1\}$, provided the same fixed choice is used in each denominator: the cross identity is unchanged. This does not negate the nonlinear Navier–Stokes equation.

Here is the exact full-flow map. In Cartesian coordinates let $O = \text{diag}(1, -1, 1)$ and define

$$u^\sharp(x, t) = Ou(Ox, t), \quad p^\sharp(x, t) = p(Ox, t), \quad f^\sharp(x, t) = Of(Ox, t).$$

The chain rule gives $\nabla u^\sharp = O(\nabla u)(Ox, t)O$, $\Delta u^\sharp = O(\Delta u)(Ox, t)$, and $(u^\sharp \cdot \nabla)u^\sharp = O((u \cdot \nabla)u)(Ox, t)$. Thus $\nabla \cdot u^\sharp = 0$ and $R_{\nu_{\text{NS}}}(u^\sharp, p^\sharp) = OR_{\nu_{\text{NS}}}(u, p)(Ox, t)$ with the same positive viscosity. In cylindrical components this is

$$(u_r^\sharp, u_\theta^\sharp, u_z^\sharp)(r, \theta, z, t) = (u_r, -u_\theta, u_z)(r, -\theta, z, t),$$

and the force transforms identically. Pressure reflects in its argument. For an axisymmetric field this reverses exactly its swirl; meridional components remain fixed. The entire momentum tensor is mapped to OTO^T , so its radial–tangential pair transforms by $(T_{r\theta}, T_{rz}) \mapsto (-T_{r\theta}, T_{rz})$ after angular averaging. Vorticity transforms as $\omega^\sharp(x, t) = \det(O)O\omega(Ox, t)$, obtained by applying the same derivative identities to the Levi-Civita formula for curl. The map is an involution, preserves the kinetic-energy integral (by $|\det O| = 1$), compact support, smoothness, and blowup magnitude along the reflected path. It preserves zero initial data when present. It proves transport of any established axisymmetric signed swirl divergence to the opposite sign; it does not itself prove an endpoint.

The source five-moment correction, auxiliary mean inverse, and infinite summation are subsequent source operations. Their full estimates are not re-proved here. The earlier draft formulas for them are withdrawn: in particular the fifth moment row is $\int (2RG\gamma_d - RV\Delta v) dR = -J_z$, without an extra outer R , and its reserved profile in a Q chart includes $(Q/q)^A$. The finite map proved above retains the two moments and all other residual terms, so it establishes neither a third shooting return nor an infinite modified viscous sequence.

14 The released result and its actual signed path

The source in this section is the supplied 166-page edition of *Finite Time Blowup for Navier–Stokes*, whose SHA-256 is

0e779481c4da40bd28d1e642e1d8ca57447d129610df28dfa5a11e9af8ae228f.

Its Theorem 1.1, on page 1, states the following result for the standard three-dimensional incompressible Navier–Stokes equation. For every fixed physical viscosity $\nu_{\text{NS}} > 0$, the source constructs

$$f \in C_c^\infty(\mathbb{R}^3 \times (0, \infty); \mathbb{R}^3), \quad u, p \in C^\infty(\mathbb{R}^3 \times [0, 1)),$$

with a fixed compact spatial support K for u and p , such that

$$\begin{aligned} \partial_t u + (u \cdot \nabla)u - \nu_{\text{NS}}\Delta u + \nabla p &= f, \quad \nabla \cdot u = 0, \quad u(\cdot, 0) = 0, \\ \sup_{t < 1} \|u(t)\|_2 &< \infty, \quad \limsup_{t \uparrow 1} \|u(t)\|_\infty = \infty. \end{aligned} \tag{41}$$

It also excludes a global smooth solution with the same force and datum and uniformly bounded kinetic energy. Its Corollary 10.6, pages 125–126, gives a periodic construction on $\mathbb{R}^3/\mathbb{Z}^3$ with

smooth time-compact force, zero initial datum and no global smooth continuation. The source identifies these with alternatives (C) and (D) in the referenced Millennium problem statement. The force is nonzero, as the source explicitly notes on page 124. These are statements of the cited source theorem; the finite algebra checks in this lane do not constitute an independent verification of the complete 166-page proof.

The source proves more than an unsigned norm divergence. In its viscosity-one construction, equations (10.20)–(10.21), page 124, give

$$\begin{aligned} A &= \frac{1}{2} + h, & 0 < h < 1/100, & & X_{\text{in}} > 0, & & e_0 > 0, \\ x_\tau &= (\sqrt{2X_{\text{in}}\tau}, 0, 0), & t_\tau &= 1 - \tau, \\ u_\theta(x_\tau, t_\tau) &= \tau^{-A}(e_0 + \epsilon(\tau)), & |\epsilon(\tau)| &\leq C\tau^{2h} \quad (0 < \tau < \tau_0). \end{aligned} \tag{42}$$

Here C, τ_0 are finite constants supplied by the source asymptotic; e_0 and X_{in} retain their source values. If $C > 0$, choose $0 < \tau_1 \leq \tau_0$ so that $C\tau_1^{2h} \leq e_0/2$; if $C = 0$, take $\tau_1 = \tau_0$. It follows directly that

$$u_\theta(x_\tau, t_\tau) \geq \frac{e_0}{2}\tau^{-A} \longrightarrow +\infty \quad (0 < \tau < \tau_1).$$

The observation point moves to the origin. This assertion does not replace that path by the fixed point $x = 0$. On the path, $q = \tau > 0$, and at its cylindrical angle $\theta = 0$ one has $e_\theta = e_2$. The geometric domain $q > 0$, the positive exponent h , and the signed velocity value are three retained parts of the same explicit formula.

15 Exact propagation at retained viscosities and orientations

For any smooth fields u, p define the physical momentum residual

$$R_{\nu_b}(u, p) = \partial_t u + (u \cdot \nabla)u - \nu_b \Delta u + \nabla p.$$

Let $\nu_b > 0$ and $\nu_t > 0$ be the base and target physical viscosities, and retain

$$\rho = \sqrt{\nu_t/\nu_b}, \quad O^T O = O O^T = I, \quad a \in \mathbb{R}^3.$$

Define the affine spatial coordinate $y = O^T(x - a)/\rho$ and the fields

$$u^\sharp(x, t) = \rho O u(y, t), \quad p^\sharp(x, t) = \rho^2 p(y, t), \quad f^\sharp(x, t) = \rho O f(y, t). \tag{43}$$

No time coordinate is changed. These formulas give a bijection on smooth field triples, with inverse

$$u(y, t) = \rho^{-1} O^T u^\sharp(a + \rho O y, t), \quad p(y, t) = \rho^{-2} p^\sharp(a + \rho O y, t), \quad f(y, t) = \rho^{-1} O^T f^\sharp(a + \rho O y, t).$$

For the Jacobian convention $(Du)_{ij} = \partial_j u_i$, differentiation gives, without dropping an orientation or viscosity,

$$\begin{aligned} Du^\sharp &= O(Du)(y, t)O^T, & \nabla \cdot u^\sharp &= (\nabla \cdot u)(y, t), \\ \partial_t u^\sharp &= \rho O \partial_t u(y, t), & (u^\sharp \cdot \nabla_x)u^\sharp &= \rho O((u \cdot \nabla_y)u)(y, t), \\ \Delta_x u^\sharp &= \rho^{-1} O \Delta_y u(y, t), & \nabla_x p^\sharp &= \rho O \nabla_y p(y, t). \end{aligned}$$

The nonlinear identity follows by multiplying $Du^\sharp$ by $u^\sharp$ and using $O^T O = I$. The Laplacian identity follows by contracting the two spatial derivative indices and using orthogonality. Since $\nu_t = \rho^2 \nu_b$, these identities prove

$$R_{\nu_t}(u^\sharp, p^\sharp) - f^\sharp = \rho O(R_{\nu_b}(u, p) - f)(y, t). \tag{44}$$

In particular the map preserves the exact equation as well as each nonzero residual with its exact factor.

For vorticity, the cross-product identity $(Ov) \times (Ow) = \det(O)O(v \times w)$ gives

$$\omega^\sharp(x, t) = \det(O)O\omega(y, t). \quad (45)$$

One proof of this identity takes its scalar product with Oz : both sides give $\det(O)\det[v, w, z]$ for every z . Applying it to the contracted first derivatives proves the vorticity formula; the factor ρ in velocity cancels the inverse factor in a derivative.

Spatial support becomes $a + \rho OK$. The Jacobian has absolute determinant ρ^3 , for both signs of $\det O$. Hence, for $1 \leq p < \infty$,

$$\|u^\sharp(t)\|_p = \rho^{1+3/p}\|u(t)\|_p, \quad \|u^\sharp(t)\|_\infty = \rho\|u(t)\|_\infty, \quad \|u^\sharp(t)\|_2^2 = \rho^5\|u(t)\|_2^2. \quad (46)$$

The Frobenius norm of $O(Du)O^T$ equals that of Du . Consequently, with integrals over the same time interval,

$$\nu_t \int \|\nabla u^\sharp\|_2^2 dt = \rho^5 \nu_b \int \|\nabla u\|_2^2 dt.$$

For arbitrary directions $v_1, \dots, v_j$, the exact force derivative is

$$D_x^j \partial_t^m f^\sharp[v_1, \dots, v_j](x, t) = \rho^{1-j} O(D_y^j \partial_t^m f)[O^T v_1, \dots, O^T v_j](y, t).$$

Thus smoothness, compact space-time support and vanishing of every derivative at a terminal point are preserved. The initial datum remains zero. A global smooth bounded-energy competitor for the transformed datum and force would, under the displayed inverse, be a competitor for the base datum and force, because its energy is multiplied by the fixed factor ρ^{-5} . This proves propagation of the source noncontinuation conclusion through the map.

16 The source positive branch and the exact negative branch

Apply the map to the source field in (42). Here $\nu_b = 1$ specifies the source object being used; physical target viscosity ν_t remains arbitrary and $\rho = \sqrt{\nu_t}$ is retained. Take $a = 0$ and the two particular orthogonal matrices

$$O_+ = I, \quad O_- = \text{diag}(1, -1, 1).$$

Both fix e_1 , so they give the same spatial observation path $x_\tau^\sharp = \rho x_\tau$ and the same time $1 - \tau$. Since $O_+ e_2 = e_2$ and $O_- e_2 = -e_2$, their exact signed values are

$$u_{\pm, \theta}^\sharp(\rho x_\tau, 1 - \tau) = \pm \rho \tau^{-A}(e_0 + \epsilon(\tau)) \longrightarrow \begin{cases} +\infty, & +, \\ -\infty, & -. \end{cases} \quad (47)$$

All support, energy, force and viscosity statements in the preceding section apply to both triples. For the negative branch the force is $\rho O_- f(O_- x / \rho, t)$. Equality of this force with the positive branch force is not needed and is not asserted.

For a general, possibly nonaxisymmetric field the reflection alone has the exact cylindrical component formula

$$(u^r, u^\theta, u^z)^\sharp(r, \theta, z, t) = (u^r, -u^\theta, u^z)(r, -\theta, z, t).$$

Indeed $O_- e_r(-\theta) = e_r(\theta)$, $O_- e_\theta(-\theta) = -e_\theta(\theta)$ and $O_- e_z = e_z$. This proves the formula even in the presence of all oscillatory corrections. For the axisymmetric background the argument $-\theta$

can be dropped because its components are independent of θ . For a velocity covariance tensor the exact Cartesian transformation is $T^\sharp(x, t) = \rho^2 O T(y, t) O^T$: expand the outer product of $\rho O w$ with itself and use linearity of the retained average. For its angularly averaged symmetric tensor, the reflection basis calculation at $\rho = 1$ gives

$$(T_{r\theta}, T_{rz})^\sharp = (-T_{r\theta}, T_{rz}).$$

Under the additional viscosity dilation its components have factor ρ^2 at the corresponding points. Thus the two covariance columns and their target transform together. Positive coefficients in a source representation $T = c_1 v_1 + c_2 v_2$, $c_i > 0$, remain the same coefficients of the transformed columns; no change of coefficient sign is required to realize the negative-swirl branch.

The source's periodic construction also has this reflection: O_- preserves $\mathbb{Z}^3$, so $[x] \mapsto [O_- x]$ is well defined on $\mathbb{R}^3/\mathbb{Z}^3$. The derivative identities apply to periodic lifts. This gives both signed orientations of the source Corollary 10.6. Arbitrary $O \in O(3)$ need not preserve this particular lattice; only the lattice-preserving reflection just specified is used for this periodic assertion.

17 Propagation into the coupled calculation

The preceding sections use the released whole-space and periodic constructions at the precise places where those source results apply. The original coupled Boussinesq work is due to Levent Alpöge and Tristan Buckmaster. This lane's retained coupled system, its finite controlled stages and its residual maps remain the separate objects defined in the preceding parts of this reader.

For every compact candidate $(u_* + w_y, p_* + \pi_y)$ in the covariance part, (44) propagates its full computed residual. Linearity of the transformation and the exact nonlinear identity give in particular

$$\Delta R_y^\sharp(x, t) = \rho O \Delta R_y(y, t).$$

Thus the finite candidate's cutoff forces, curl errors and covariance remainders all have an exact transformed counterpart. A nonzero remainder is carried to a nonzero remainder, because the map is invertible. The source's residual flatness at the singular point belongs to its completed local source construction; its extended smooth force need not vanish on the entire time-one slice. The next section calculates the surviving moments of the actual finite candidate, so that propagation continues through its computed terms rather than assuming that its different coupled coefficients inherit the source's infinite estimates.

The earlier suggestion of a negative blowout imposes no condition on the programme. Formula (42) is the source-selected sign; formula (47) proves the exact relation of its two orientations. Both formulas retain a positive physical viscosity and the source coordinate domain $q > 0$.

18 The two moments of the actual compact curl increment

The objects in this calculation are the finite smooth source state (u_*, p_*) and the actual added curl field $v = w_y$ with its selected pressure $\pi = \pi_y$ from the signed covariance bridge. Their physical residual increment is

$$\Delta R = \partial_t v + (u_* \cdot \nabla) v + (v \cdot \nabla) u_* + (v \cdot \nabla) v - \nu_{\text{NS}} \Delta v + \nabla \pi. \quad (48)$$

The physical viscosity ν_{NS} is unchanged. Both velocities are divergence free: the source state uses its actual divergence-free background and curl increments, and v is the curl of its specified compact potential. No residual equation for u_* is required in (48); its residual has been subtracted exactly.

All radial supports involving v or π lie in a fixed compact subinterval of $(0, \infty)$ on the compact space–time patch at issue.

Here the source extended field is indispensable. Its absolute auxiliary coordinate is $Y \in \mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$, and physical evaluation uses precisely source (6.2)–(6.4):

$$\begin{aligned} J_g &= \begin{pmatrix} 3 & 1 \\ 1 & 5 \end{pmatrix}, \quad \Lambda_g = 4 - \sqrt{2}, \quad T_g = 4 + \sqrt{2}, \quad b_g = \sqrt{2} - 1, \\ \mathbf{v}_r &= (1, -b_g), \quad \mathbf{v}_t = (b_g, 1), \quad \rho_g = \frac{\log \Lambda_g}{\log T_g}, \quad \kappa_s = 10^{-5}, \quad d_r = 2((1+h)\rho_g - h\kappa_s), \\ Y(r, t) &= \mathbf{v}_r r^{d_r} + \mathbf{v}_t t \pmod{\mathbb{Z}^2}, \quad \mathcal{D}_r = \partial_r + d_r r^{d_r-1} \mathbf{v}_r \cdot \partial_Y, \quad \mathcal{D}_t = \partial_t + \mathbf{v}_t \cdot \partial_Y. \end{aligned} \quad (49)$$

These are the source’s original quantities, not the coupled physical frequency or diffusivity. Angular and axial physical derivatives are ∂_θ and ∂_z , because $Y(r, t)$ depends on neither variable. Every field below is first differentiated using these lifted physical derivatives. A tilde is suppressed for this extended representation. Evaluation at (49) recovers the physical field and its physical derivatives by the chain rule.

The incompressibility needed below holds at every independent Y . Indeed the actual finite source construction retains the form $u_* = \text{curl}_{\mathcal{D}} \mathcal{A}_* + B_* e_\theta$ with $\partial_\theta B_* = 0$, and the added field is $v = \text{curl}_{\mathcal{D}} \mathcal{A}_y$. The retained source potentials here include its finite mean and wave curl increments. The lifted derivatives in (49), together with ∂_θ and ∂_z , commute, and $\mathcal{D}_r r = 1$. Substitution in the cylindrical curl and divergence formulas therefore gives $\text{div}_{\mathcal{D}} \text{curl}_{\mathcal{D}} \mathcal{A} = 0$ identically in (r, θ, z, t, Y) : each mixed derivative cancels its opposite, including the derivative of r^{-1} . Also $\text{div}_{\mathcal{D}}(B_* e_\theta) = r^{-1} \partial_\theta B_* = 0$. Thus extended incompressibility is proved from the actual retained potential representation, rather than inferred from its restriction to the physical phase map.

Use the normalized averages

$$\langle a \rangle_\theta = \frac{1}{2\pi} \int_0^{2\pi} a \, d\theta, \quad \langle a \rangle_Y = \int_{\mathbb{T}^2} a \, dY, \quad \langle a \rangle = \langle \langle a \rangle_\theta \rangle_Y, \quad \int_{\mathbb{T}^2} 1 \, dY = 1.$$

They act on cylindrical components before physical evaluation, exactly as source (3.7)–(3.9). All claimed integral identities below concern these specified averages; the exact relation to evaluated moments is also proved below.

18.1 The actual harmonic curl has zero angular mean

For each retained source harmonic, its extended potential and pressure have cylindrical components of the form

$$C_j(r, z, t, Y) e^{i(n_j \theta + \psi_j(r, z, t, Y))}, \quad P_j(r, z, t, Y) e^{i(n_j \theta + \psi_j(r, z, t, Y))}, \quad n_j = k_{\beta_j} m_j p_j \in \mathbb{Z} \setminus \{0\}.$$

The coefficients include the actual cutoff χ , frame B , map L , coupled solution $y = (\Theta, \Omega)^T$, band powers and pressure formula. Their cylindrical components are independent of θ . In particular, no sign of Θ , Ω or the real part of a harmonic is prescribed. For a vector potential $\mathcal{A}$ the exact cylindrical curl is

$$(\text{curl } \mathcal{A})_r = r^{-1} \partial_\theta \mathcal{A}_z - \partial_z \mathcal{A}_\theta, \quad (\text{curl } \mathcal{A})_\theta = \partial_z \mathcal{A}_r - \mathcal{D}_r \mathcal{A}_z, \quad (\text{curl } \mathcal{A})_z = (\mathcal{D}_r + r^{-1}) \mathcal{A}_\theta - r^{-1} \partial_\theta \mathcal{A}_r.$$

These formulas include the derivatives of the cylindrical basis. Each displayed operation retains the same factor $e^{in_j \theta}$: the coefficients of $\mathcal{D}_r$ are independent of θ , and ∂_θ multiplies that factor by

in_j . Integration of the factor over the angular circle gives zero. Taking real parts and the actual finite harmonic sum proves

$$\langle v_r \rangle_\theta = \langle v_\theta \rangle_\theta = \langle v_z \rangle_\theta = \langle \pi \rangle_\theta = 0 \quad \text{at every } (r, z, t, Y). \quad (50)$$

This includes every curl and cutoff correction, not merely the principal amplitude. The same identities hold after physical evaluation because $Y(r, t)$ is independent of θ .

For every smooth periodic coefficient a , integration of its auxiliary derivatives gives

$$\langle \mathcal{D}_r a \rangle = \partial_r \langle a \rangle, \quad \langle \mathcal{D}_t a \rangle = \partial_t \langle a \rangle, \quad \langle \partial_z a \rangle = \partial_z \langle a \rangle, \quad \langle \partial_\theta a \rangle = 0. \quad (51)$$

There is no omitted variable coefficient in this assertion: $d_r r^{d_r-1} \mathbf{v}_r$ is independent of Y . Applying the first identity again proves $\langle \mathcal{D}_r^2 a \rangle = \partial_r^2 \langle a \rangle$. Explicitly, if $c(r) = d_r r^{d_r-1} \mathbf{v}_r$, then

$$\mathcal{D}_r^2 a = \partial_r^2 a + 2c \cdot \partial_Y \partial_r a + c' \cdot \partial_Y a + (c \cdot \partial_Y)^2 a;$$

each of the three additional terms has zero auxiliary average.

18.2 Exact removal of the linear and mean-flow terms

Put

$$\bar{u}_* := \langle u_* \rangle_\theta, \quad w_* := u_* - \bar{u}_*.$$

The field $\bar{u}_*$ may retain auxiliary dependence; none is discarded. Since it is independent of θ , (50) gives

$$\langle \bar{u}_* \otimes v + v \otimes \bar{u}_* \rangle = 0.$$

Define the actual symmetric cross-and-quadratic tensor

$$S := \langle w_* \otimes v + v \otimes w_* + v \otimes v \rangle. \quad (52)$$

Products of all original source waves with the added curl field are retained in this formula, including old and new curl remainders.

For two divergence-free fields a, b , Cartesian component expansion gives

$$\nabla \cdot (a \otimes b + b \otimes a) = (b \cdot \nabla) a + (a \cdot \nabla) b, \quad \nabla \cdot (v \otimes v) = (v \cdot \nabla) v.$$

For example $\sum_j \partial_j (a_i b_j) = \sum_j b_j \partial_j a_i + a_i \sum_j \partial_j b_j$; the last term is zero. The lifted derivatives obey the same product rule, so the identities hold on the extended domain before evaluation. The three advection terms in (48) are therefore exactly the divergence of $u_* \otimes v + v \otimes u_* + v \otimes v$.

For completeness, set $\mathcal{L}_0 = \mathcal{D}_r^2 + r^{-1} \mathcal{D}_r + r^{-2} \partial_\theta^2 + \partial_z^2$. The vector Laplacian and pressure gradient in the two components are

$$(\Delta v)_\theta = (\mathcal{L}_0 - r^{-2}) v_\theta + 2r^{-2} \partial_\theta v_r, \quad (\Delta v)_z = \mathcal{L}_0 v_z, \quad (\nabla \pi)_\theta = r^{-1} \partial_\theta \pi, \quad (\nabla \pi)_z = \partial_z \pi.$$

Equations (50)–(51) prove separately

$$\langle \mathcal{D}_t v \rangle_{\text{tan}} = 0, \quad -\nu_{\text{NS}} \langle \Delta v \rangle_{\text{tan}} = 0, \quad \langle \nabla \pi \rangle_{\text{tan}} = 0. \quad (53)$$

Thus viscosity and pressure were included before averaging; their vanishing here is a proved consequence of the actual carriers.

If T is symmetric, differentiation of $e_r = (\cos \theta, \sin \theta, 0)$, $e_\theta = (-\sin \theta, \cos \theta, 0)$ gives

$$\begin{aligned}(\nabla \cdot T)_\theta &= (\mathcal{D}_r + 2/r)T_{r\theta} + r^{-1}\partial_\theta T_{\theta\theta} + \partial_z T_{z\theta}, \\(\nabla \cdot T)_z &= (\mathcal{D}_r + 1/r)T_{rz} + r^{-1}\partial_\theta T_{\theta z} + \partial_z T_{zz}.\end{aligned}$$

Indeed $\partial_\theta e_r = e_\theta$ and $\partial_\theta e_\theta = -e_r$ supply the two radial connection terms in the first line and the single radial connection term in the second. Averaging these formulas and using (51), (52), and (53) proves the exact residual identities

$$F_2 := \langle \Delta R_\theta \rangle = (\partial_r + 2/r)S_{r\theta} + \partial_z S_{z\theta}, \quad F_1 := \langle \Delta R_z \rangle = (\partial_r + 1/r)S_{rz} + \partial_z S_{zz}. \quad (54)$$

In particular every mean-flow cross term vanishes as the divergence of its zero averaged tensor, even when the mean depends on Y .

18.3 The surviving axial fluxes and their signs

Define the two moments with the original physical radial measure:

$$M_2(z, t) = \int_0^\infty r^2 F_2(r, z, t) dr, \quad M_1(z, t) = \int_0^\infty r F_1(r, z, t) dr. \quad (55)$$

Their exact values are

$$M_2 = \partial_z J_2, \quad J_2 = \int_0^\infty r^2 \langle w_{*,z} v_\theta + v_z w_{*,\theta} + v_z v_\theta \rangle dr, \quad (56)$$

$$M_1 = \partial_z J_1, \quad J_1 = \int_0^\infty r \langle 2w_{*,z} v_z + v_z^2 \rangle dr. \quad (57)$$

To prove them, multiply (54) by the respective weight: $r^2(\partial_r + 2/r)S_{r\theta} = \partial_r(r^2 S_{r\theta})$ and $r(\partial_r + 1/r)S_{rz} = \partial_r(r S_{rz})$. The weighted boundary values at both ends of $(0, \infty)$ are zero, because each entry of S contains at least one compactly supported factor v . Differentiation under the radial integral is valid by the same compact support and smoothness. This proves (56)–(57) including their signs and factors.

There is no pointwise cancellation of the axial derivatives in these formulas. Since the added field has compact axial support, J_1 and J_2 have compact axial support and

$$\int_{\mathbb{R}} M_e(z, t) dz = 0 \quad (e = 1, 2). \quad (58)$$

This follows by evaluating the boundary values of J_e at the two axial ends. It does not assert $M_e(z, t) = 0$ at an individual axial point. The compact radial primitive in the covariance bridge must therefore retain its $b_e M_e$ terms with these actual M_e .

At fixed source state, replacing the added field and pressure by their negatives sends the old-wave cross terms in (56)–(57) to their negatives and leaves the quadratic terms unchanged. Thus negating an amplitude is not a sign reversal of either full moment. Under the full physical reflection $O = \text{diag}(1, -1, 1)$, applied to the source state and added field together as $a^\sharp(x, t) = Oa(Ox, t)$, the cylindrical radial and axial components retain their signs and the angular components reverse sign at $(r, -\theta, z, t)$. Changing the angular integration variable therefore proves

$$J_2^\sharp = -J_2, \quad M_2^\sharp = -M_2, \quad J_1^\sharp = J_1, \quad M_1^\sharp = M_1.$$

An explicit transverse-curl example shows why compactness and nonzero angular frequency alone do not force either moment to vanish. This example tests precisely those structural properties, and is not a replacement for the retained source state. Let H be any nonzero real $C_c^\infty((r_-, r_+))$ function with $0 < r_- < r_+$, let $g \in C_c^\infty(\mathbb{R})$ be nonconstant, let n be a nonzero integer, and put $f = rH'$. Use zero old state and the real potential

$$\mathcal{A}_r = fg \sin(n\theta), \quad \mathcal{A}_\theta = 0, \quad \mathcal{A}_z = Hg \cos(n\theta).$$

It has the required transverse representation: $\phi = n\theta$, $\xi = (n/r)e_\theta$, $C = (-ifg, 0, Hg)$ in cylindrical components, $a = i\xi \times C$, and $C = i\xi \times a/|\xi|^2$ because $\xi \cdot C = 0$. The curl is exactly

$$v_r = -\frac{nHg}{r} \sin(n\theta), \quad v_\theta = fg' \sin(n\theta) - H'g \cos(n\theta), \quad v_z = -\frac{nfg}{r} \cos(n\theta).$$

Consequently

$$M_2 = ngg' \int_0^\infty r^2 (H')^2 dr, \quad M_1 = n^2 gg' \int_0^\infty r (H')^2 dr.$$

For example $g(z) = \exp(-1/(1-z^2))$ on $|z| < 1$, extended by zero, has $g(z)g'(z) \neq 0$ at every $0 < |z| < 1$. Both integrals are strictly positive: a compactly supported smooth function with $H' = 0$ everywhere would be constant and hence zero. Thus both moments can be nonzero, with their displayed signed values, within the exact transverse-curl architecture.

18.4 Bounds in the actual fields and original chart units

At fixed (z, t) set

$$\|a\|_e^2 = \int_0^\infty r^e \langle |a|^2 \rangle dr, \quad e = 1, 2.$$

When an old-field norm is used below it can be restricted to the enclosing radial support of v and $\partial_z v$. This avoids imposing any decay condition on an unused exterior part of the old field. The following component bounds contain only actual fields:

$$|J_2| \leq \|w_{*,z}\|_2 \|v_\theta\|_2 + \|w_{*,\theta}\|_2 \|v_z\|_2 + \|v_z\|_2 \|v_\theta\|_2, \quad (59)$$

$$|J_1| \leq 2\|w_{*,z}\|_1 \|v_z\|_1 + \|v_z\|_1^2, \quad (60)$$

$$|M_2| \leq \|\partial_z w_{*,z}\|_2 \|v_\theta\|_2 + \|w_{*,z}\|_2 \|\partial_z v_\theta\|_2 + \|\partial_z v_z\|_2 \|w_{*,\theta}\|_2 \\ + \|v_z\|_2 \|\partial_z w_{*,\theta}\|_2 + \|\partial_z v_z\|_2 \|v_\theta\|_2 + \|v_z\|_2 \|\partial_z v_\theta\|_2, \quad (61)$$

$$|M_1| \leq 2(\|\partial_z w_{*,z}\|_1 \|v_z\|_1 + \|w_{*,z}\|_1 \|\partial_z v_z\|_1 + \|v_z\|_1 \|\partial_z v_z\|_1). \quad (62)$$

Each is the Cauchy–Schwarz inequality on the product measure $r^e dr d\theta / (2\pi) dY$, after applying the product rule in (56)–(57). For example, the derivative of $2w_{*,z}v_z + v_z^2$ is $2(\partial_z w_{*,z})v_z + 2w_{*,z}\partial_z v_z + 2v_z\partial_z v_z$, proving every coefficient in (62). For a (z, t) multi-index I , every further derivative is exactly obtained by replacing each product ab in J_e by

$$\partial^{I+(1,0)}(ab) = \sum_{K \leq I+(1,0)} \binom{I+(1,0)}{K} (\partial^K a) (\partial^{I+(1,0)-K} b)$$

inside its original radial integral. This identity follows inductively from the one-derivative product rule, and gives the corresponding finite-order moment bound by applying the same Cauchy–Schwarz inequality separately to every displayed summand.

The field v in these bounds is not an abstract replacement amplitude. For each actual band and harmonic write

$$t_j = \chi_j B_j L_j y_j, \quad c_j = \frac{i n_{\Phi,j} \times t_j}{k_{\beta_j} m_j |n_{\Phi,j}|^2}, \quad b_j = t_j + \text{curl}_{*,\text{coeff}} c_j, \\ v = \sum_j \text{Re}(Q_{\beta_j}^{-A} b_j e^{ik_{\beta_j} m_j \Phi_j}), \quad A = \frac{1}{2} + h, \quad D = \frac{1}{2} - h. \quad (63)$$

Here $\text{curl}_{*,\text{coeff}}$ differentiates the chart coefficient, including the cutoff, auxiliary chain rule and cylindrical basis, but not the displayed exponential. This is exactly the remainder of the physical potential $Q_{\beta_j}^{1/2-A} c_j e^{ik_{\beta_j} m_j \Phi_j}$: physical curl introduces the factor $Q_{\beta_j}^{-1/2}$, and differentiating the exponential produces t_j by the transverse triple-product identity. Explicitly, on the absolute auxiliary torus in a fixed band,

$$\mathcal{D}_R = \partial_R + \sqrt{Q} d_r r^{d_r-1} \mathbf{v}_r \cdot \partial_Y, \quad r = \sqrt{Q} R, \quad \mathcal{D}_Z^* = \varepsilon \partial_Z.$$

For the actual coefficient $c = (c_r, c_\theta, c_z)$, which has $\partial_\theta c = 0$, the operator is exactly

$$\text{curl}_{*,\text{coeff}} c = (-\varepsilon \partial_Z c_\theta, \varepsilon \partial_Z c_r - \mathcal{D}_R c_z, (\mathcal{D}_R + R^{-1}) c_\theta).$$

Coefficients originally on a band auxiliary torus are pulled back to the absolute torus before applying this formula; this retains the integer-covering chain derivative in $\partial_Y c$. In the same fixed chart the exact axial derivative is

$$\partial_z v = \sum_j \text{Re} \left[Q_{\beta_j}^{-A-D} (\partial_{Z_j} b_j + ik_{\beta_j} m_j (\partial_{Z_j} \Phi_j) b_j) e^{ik_{\beta_j} m_j \Phi_j} \right]. \quad (64)$$

The band power is fixed during chart differentiation and $\partial_z = Q_{\beta_j}^{-D} \partial_{Z_j}$; auxiliary phase evaluation adds no axial derivative because of (49). Thus (63)–(64) substitute the actual retained pair and all its curl terms into the bounds. For example, writing $\|b\|_{e,\beta}^2 = \int R^e \langle |b|^2 \rangle dR$ in that band, the triangle inequality and $|\text{Re } z| \leq |z|$ give

$$\|v\|_e \leq \sum_j Q_{\beta_j}^{(e+1)/4-A} \|b_j\|_{e,\beta_j}, \quad (65)$$

$$\|\partial_z v\|_e \leq \sum_j Q_{\beta_j}^{(e+1)/4-A-D} \|\partial_{Z_j} b_j + ik_{\beta_j} m_j (\partial_{Z_j} \Phi_j) b_j\|_{e,\beta_j}. \quad (66)$$

The exponent $(e+1)/4$ is exactly the square root of the radial measure factor $r^e dr = Q^{(e+1)/2} R^e dR$. Distinct bands and harmonics remain distinct in these sums. No limiting estimate as $q \downarrow 0$ has been inferred from the finite compact-patch norms.

For a single fixed chart, put $F_e = Q^{-2A-1/2} E_e^*$ as in the covariance bridge and $S^* = Q^{2A} S$. Direct change of radial variable gives

$$M_e = Q^{e/2-2A} M_e^*, \quad M_e^* = \int R^e E_e^* dR, \quad M_e^* = \varepsilon \partial_Z \int R^e S_{z,\tau_e}^* dR, \quad \varepsilon = Q^h,$$

where $\tau_2 = \theta$ and $\tau_1 = z$. Indeed the axial flux has factor $Q^{(e+1)/2-2A}$, its axial derivative contributes Q^{-D} , and dividing by $Q^{e/2-2A}$ leaves $Q^{1/2-D} = Q^h$. This proves the precise relation to the source chart moment without altering its axial anisotropy.

18.5 Exact relation to moments after phase evaluation

Let

$$\mathcal{S} = \langle w_* \otimes v + v \otimes w_* + v \otimes v \rangle_\theta, \quad \mathcal{S}^\circ = \mathcal{S} - \langle \mathcal{S} \rangle_Y.$$

Thus $\langle \mathcal{S} \rangle_Y = S$ and $\langle \mathcal{S}^\circ \rangle_Y = 0$. For the evaluated physical fields, denote the angular-only residual moments by M_e^{ev} . The proof of (50) holds after evaluation. The physical divergence, vector Laplacian and gradient identities therefore prove, without any auxiliary averaging,

$$M_e^{\text{ev}} = \partial_z \int_0^\infty r^e \mathcal{S}_{z,\tau_e}(r, z, t, Y(r, t)) dr.$$

The physical radial boundary terms still vanish by compact support; the chain rule is included in the physical radial derivative before that integration. Subtracting (56)–(57) proves the exact comparison

$$M_e^{\text{ev}} - M_e = \partial_z \int_0^\infty r^e \mathcal{S}_{z,\tau_e}^\circ(r, z, t, Y(r, t)) dr. \quad (67)$$

For example its absolute value is bounded by $\int r^e \sup_Y |\partial_z \mathcal{S}_{z,\tau_e}^\circ| dr$, because $Y(r, t)$ is independent of z . Zero auxiliary Haar average does not make the right side of (67) zero at the physical phase. The displayed map supplies precisely the remaining auxiliary-oscillation flux; it is not silently identified with the source fast mean.

Part II

Original third-growth derivatives

19 Time derivatives of the actual signed third growth

This calculation is on the actual interval already constructed in *The actual third growth target on the moving second hold* and *Exact third growth on the moving second holding state*. It gives quantitative derivatives of that trajectory for the source-frame map. All derivatives in this section are with respect to coupled physical time t_c , and $\tau = t_c - t_b$ denotes elapsed coupled time. The notation does not use the cylindrical spatial radius for elapsed time.

19.1 Original data, evolving parent, and proved interval

Retain the source parameters

$$\begin{aligned} \nu &= \sqrt{A_0}, & \mu &= \lambda_0 = \lceil \sqrt{A_0}/2 \rceil, & K_j &= \mu \hat{\lambda}_j, \\ S_3 &= \frac{A_0}{\mu} \hat{\lambda}_3^{-k_3-6}, & P_3^* &= \frac{A_0}{\mu} \hat{\lambda}_3^{-7/8} = S_3 e^{L_3}. \end{aligned} \quad (68)$$

The physical common diffusivity is $d > 0$; it is distinct from ν . Use the actual second return t_b , its values $h_b, A_{1,b}, P_{2,b}$, and the original angles s_*, s_2, s_3 . Put

$$\begin{aligned} a &= \sin s_2, & \chi_b &= \sqrt{h_b^2 + a^2}, & \beta_b &= \arctan(a/h_b), & b &= \sin(s_3 + \beta_b), & g_b &= \cos(s_3 + \beta_b), \\ c &= bh_b - ag_b = \chi_b \sin s_3, & S_* &= A_0 \sin s_*, & C_* &= A_0 \cos s_*. \end{aligned} \quad (69)$$

Here $a, b, c, S_*, C_* > 0$, and the letter b is the original oriented geometric constant, whereas b_3 below is an amplitude coefficient. The moving parent is exactly

$$\begin{aligned} \chi^2 &= h^2 + a^2, & \dot{h} &= a\Omega_1, & \dot{\Omega}_1 &= aA_1/\chi - dK_1^2\Omega_1, \\ \dot{A}_1 &= -S_*\Omega_1 - dK_1^2A_1, & \dot{P}_2 &= -dK_2^2\chi^2P_2, & \dot{g} &= b\Omega_1, \\ g &= g_b + (b/a)(h - h_b), & N_3 &= b(A_1 + C_*) + gS_* + cK_2P_2, & R_3 &= g^2 + b^2. \end{aligned} \quad (70)$$

The retained feedback is $\dot{\alpha} = -a^2\Omega_1/\chi^2$ and $\Omega_2 = 0$. In particular the older temperature and first vorticity are not frozen. The cancellation in the complete numerator is

$$\dot{N}_3 = -dE_3, \quad E_3 = bK_1^2A_1 + cK_2^3\chi^2P_2. \quad (71)$$

Write $N_i = N_3(t_b)$, $a_i = N_i/K_3$, $b_i = K_3 \sin s_3$, $\gamma_i = \sqrt{a_i b_i}$ and $u_i = \sqrt{b_i/a_i}$. The exact new equations, including their original signed seed, are

$$\begin{aligned} y &= \begin{pmatrix} \Theta_3 \\ \Omega_3 \end{pmatrix}, & \dot{y} &= C_3 y, & C_3 &= \begin{pmatrix} -\lambda_3 & a_3 \\ b_3 & -\lambda_3 \end{pmatrix}, \\ a_3 &= \frac{N_3}{K_3 R_3}, & b_3 &= \frac{K_3 c}{\chi}, & \lambda_3 &= dK_3^2 R_3, & y(t_b) &= -S_3 \begin{pmatrix} 1 \\ u_i \end{pmatrix}. \end{aligned} \quad (72)$$

There are no new amplitude controls in this growth pair: $c_\vartheta = c_\omega = 0$. The activation of the compact wave is a separate original factor $\eta(\Gamma_3\tau)$; it does not replace (72) by a forced coefficient trajectory.

For clarity, the following facts about this actual interval were proved in the third-threshold construction. They identify the trajectory to which the estimates below apply:

$$\begin{aligned}
I_3 &= [t_b, t_3^a], \quad t_3^a - t_b = \tau_a < 14/\sigma_1, \quad P_3 = -\Theta_3 > 0, \quad P_3(t_b) = S_3, \quad P_3(t_3^a) = P_3^*, \\
u &= \Omega_3/\Theta_3, \quad \frac{u_i}{\sqrt{2}} \leq u \leq 2u_i, \quad \frac{a_i}{4} \leq a_3 \leq a_i, \quad \frac{b_i}{2} \leq b_3 \leq b_i, \\
1 &\leq R_3 \leq \bar{R} := \left(\frac{265}{256}\right)^2 < 2, \quad 1 \leq \frac{\chi}{\chi_b} \leq \frac{259}{256}, \quad 0 < dK_3^2 \leq \frac{\gamma_i}{16\sqrt{2}}, \\
\frac{\gamma_i}{8\sqrt{2}} &\leq (\log P_3)' \leq 2\gamma_i, \quad 0 < A_1 \leq A_{1,b}, \quad 0 < P_2 \leq P_{2,b}, \quad 0 < \Omega_1 < 9\sigma_1, \\
h_b &\leq h \leq h_b + 144a, \quad g_b \leq g \leq g_b + 144b, \quad \frac{1}{2}N_i \leq N_3 \leq N_i, \quad 1 \leq \chi^2 < 10.
\end{aligned} \tag{73}$$

The value of d is precisely the previously selected range $0 < d \leq d_{3,\max}$, retaining its K_3^{-2} dependence. Thus (73) does not assert a new unrestricted viscosity range or assume a further return.

19.2 Signed amplitudes and first derivatives with the diffusion retained

Theorem 1. *For every $t_c \in I_3$, both $\Theta_3(t_c)$ and $\Omega_3(t_c)$ are negative and strictly decreasing. Their exact component rates are*

$$\dot{\Theta}_3 = -P_3(a_3u - \lambda_3), \quad \dot{\Omega}_3 = -P_3(b_3 - \lambda_3u). \tag{74}$$

They obey the bounds

$$\begin{aligned}
S_3 e^{\gamma_i \tau / (8\sqrt{2})} &\leq P_3(t_c) \leq \min\{P_3^*, S_3 e^{2\gamma_i \tau}\}, \\
\frac{u_i}{\sqrt{2}} P_3 &\leq |\Omega_3| \leq 2u_i P_3, \\
\left(\frac{\gamma_i}{4\sqrt{2}} - dK_3^2 \bar{R}\right) P_3 &\leq |\dot{\Theta}_3| \leq 2\gamma_i P_3, \\
u_i \left(\frac{\gamma_i}{2} - 2dK_3^2 \bar{R}\right) P_3 &\leq |\dot{\Omega}_3| \leq \gamma_i u_i P_3.
\end{aligned} \tag{75}$$

In particular both lower derivative bounds are positive, and

$$\begin{aligned}
P_3 \sqrt{1 + u_i^2/2} &\leq |y|_2 \leq P_3 \sqrt{1 + 4u_i^2} \leq P_3^* \sqrt{1 + 4u_i^2}, \\
|\dot{y}|_2 &\leq \gamma_i P_3 \sqrt{4 + u_i^2} \leq \gamma_i P_3^* \sqrt{4 + u_i^2}.
\end{aligned} \tag{76}$$

The original beginning and attained endpoint have the exact derivatives

$$\begin{aligned}
\dot{y}(t_b) &= -S_3(\gamma_i - dK_3^2) \begin{pmatrix} 1 \\ u_i \end{pmatrix}, \\
y(t_3^a) &= -P_3^* \begin{pmatrix} 1 \\ u(t_3^a) \end{pmatrix}, \\
\dot{y}(t_3^a) &= -P_3^* \begin{pmatrix} a_3(t_3^a)u(t_3^a) - dK_3^2 R_3(t_3^a) \\ b_3(t_3^a) - dK_3^2 R_3(t_3^a)u(t_3^a) \end{pmatrix}.
\end{aligned} \tag{77}$$

Proof. The signs and quotient in (73) give $y = -P_3(1, u)^T$. Substituting this expression in the exact pair proves (74). Integrating the already proved logarithmic rate from t_b and using the attained

endpoint and strict increase of P_3 gives the first line of (75). The quotient box gives its second line. Since $a_i u_i = \gamma_i$ and $b_i = \gamma_i u_i$, direct multiplication of the positive boxes gives

$$\frac{\gamma_i}{4\sqrt{2}} \leq a_3 u \leq 2\gamma_i, \quad \frac{\gamma_i u_i}{2} \leq b_3 \leq \gamma_i u_i, \quad 0 < \lambda_3 \leq dK_3^2 \bar{R}.$$

Consequently the displayed lower bound for $a_3 u - \lambda_3$ follows. It is at least $\gamma_i/(8\sqrt{2}) > 0$, because $\bar{R} < 2$ and $dK_3^2 \leq \gamma_i/(16\sqrt{2})$. Similarly,

$$b_3 - \lambda_3 u \geq u_i(\gamma_i/2 - 2dK_3^2 \bar{R}) > \gamma_i u_i(1/2 - 1/(4\sqrt{2})) > \gamma_i u_i/4 > 0.$$

Thus both derivatives in (74) are strictly negative. Dropping their nonnegative diffusion losses only for the purpose of an upper bound gives the two upper derivative estimates; the exact rates continue to contain the losses. Squaring the two amplitude and two derivative bounds, adding them, and taking the positive square root proves (76). Finally $R_3(t_b) = 1$, $u(t_b) = u_i$, $a_3(t_b) = a_i$ and $b_3(t_b) = b_i$ give the first endpoint identity. Substitution of the proved target value gives the other two. In particular no later endpoint was substituted for the actual $u(t_3^a)$. $\square$

19.3 Explicit coefficient rates

Set $\bar{H} = h_b + 144a$, $\bar{G} = g_b + 144b$ and $\bar{W} = 9\sigma_1$. Differentiating the complete coefficients gives

$$\begin{aligned} \dot{a}_3 &= -a_3 \left(\frac{dE_3}{N_3} + \frac{2bg\Omega_1}{R_3} \right), \\ \dot{b}_3 &= -b_3 \frac{ah\Omega_1}{\chi^2}, \quad \dot{\lambda}_3 = 2dK_3^2 bg\Omega_1. \end{aligned} \tag{78}$$

Every quantity on the right comes from the moving parent. In particular the exact numerator derivative (71) is used before any upper estimate. Because $K_1 < K_2$, $\chi^2 < 10$ and all summands of N_3 are positive, $E_3 \leq 10K_2^2 N_3$. Thus

$$\begin{aligned} |\dot{a}_3| &\leq a_i(10dK_2^2 + 2b\bar{G}\bar{W}), \quad |\dot{b}_3| \leq b_i a \bar{H} \bar{W} \chi_b^{-2}, \\ |\dot{\lambda}_3| &\leq 2dK_3^2 b \bar{G} \bar{W}. \end{aligned} \tag{79}$$

These formulas also display the distinct inherited K_2^2 loss and new K_3^2 loss. They do not replace either by the source scaling parameter ν .

19.4 A closed all-orders derivative calculation

Here are finite recurrences whose values are explicit upper bounds from the actual birth data and the already proved interval. They use ordinary derivatives, with no change to physical time or amplitudes. For integers $n \geq 0$ and $m \geq 2$ define the product expression

$$\mathcal{P}_n(X^{(1)}, \dots, X^{(m)}) = \sum_{j_1 + \dots + j_m = n} \frac{n!}{j_1! \dots j_m!} X_{j_1}^{(1)} \dots X_{j_m}^{(m)}.$$

This notation is a finite sum, and repeated arguments mean repeated factors. Define the nonnegative arrays H, W, A, P, Q, G, N, J initially by

$$H_0 = \bar{H}, \quad W_0 = \bar{W}, \quad A_0^{\text{bd}} = A_{1,b}, \quad P_0 = P_{2,b}, \quad Q_0 = \chi_b^{-1}, \quad G_0 = \bar{G}, \quad N_0 = N_i, \quad J_0 = 1. \tag{80}$$

In the following recurrences the array A has entries A_n^{bd} ; the superscript distinguishes its zeroth entry from the unchanged physical constant A_0 in (68). Write

$$X_n = \mathcal{P}_n(H, H) + a^2 \mathbf{1}_{n=0}.$$

Having computed all eight arrays up to index n , compute index $n + 1$ by the explicit formulas

$$\begin{aligned} H_{n+1} &= aW_n, \\ W_{n+1} &= a\mathcal{P}_n(A, Q) + dK_1^2 W_n, \\ A_{n+1}^{\text{bd}} &= S_* W_n + dK_1^2 A_n^{\text{bd}}, \\ P_{n+1} &= dK_2^2 \mathcal{P}_n(X, P), \\ Q_{n+1} &= a\mathcal{P}_n(H, W, Q, Q, Q), \\ G_{n+1} &= bW_n, \\ N_{n+1} &= dbK_1^2 A_n^{\text{bd}} + dcK_2^3 \mathcal{P}_n(X, P), \\ J_{n+1} &= 2b\mathcal{P}_n(G, W, J, J). \end{aligned} \tag{81}$$

For the phase-length array use

$$R_0^{\text{bd}} = \bar{R}, \quad R_n^{\text{bd}} = \mathcal{P}_n(G, G) \quad (n \geq 1).$$

The coefficient arrays are

$$\begin{aligned} \mathfrak{A}_0 &= a_i, \quad \mathfrak{B}_0 = b_i, \quad \mathfrak{D}_0 = dK_3^2 \bar{R}, \\ \mathfrak{A}_n &= K_3^{-1} \mathcal{P}_n(N, J), \quad \mathfrak{B}_n = K_3 c Q_n, \quad \mathfrak{D}_n = dK_3^2 R_n^{\text{bd}} \quad (n \geq 1). \end{aligned} \tag{82}$$

No optimization of these finite sums is needed for their validity. In particular $\mathfrak{A}_1$ may be replaced by the smaller of its value above and the first bound in (79), and similarly for the other two first coefficient derivatives.

The amplitude arrays start with the actual target and the sharper first-derivative bounds already proved:

$$\begin{aligned} T_0 &= P_3^*, \quad O_0 = 2u_i P_3^*, \quad T_1 = 2\gamma_i P_3^*, \quad O_1 = \gamma_i u_i P_3^*, \\ T_{n+1} &= \sum_{j=0}^n \binom{n}{j} (\mathfrak{D}_j T_{n-j} + \mathfrak{A}_j O_{n-j}), \quad n \geq 1, \\ O_{n+1} &= \sum_{j=0}^n \binom{n}{j} (\mathfrak{B}_j T_{n-j} + \mathfrak{D}_j O_{n-j}), \quad n \geq 1. \end{aligned} \tag{83}$$

The choice to begin these recurrences at $n = 1$ preserves the stronger first-derivative estimates instead of substituting a larger matrix-norm bound.

Theorem 2. *For every integer $n \geq 0$ the finite arrays above satisfy on I_3*

$$\begin{aligned} |\partial_{t_c}^n \Theta_3| &\leq T_n, \quad |\partial_{t_c}^n \Omega_3| \leq O_n, \quad |\partial_{t_c}^n y|_2 \leq \sqrt{T_n^2 + O_n^2}, \\ |\partial_{t_c}^n a_3| &\leq \mathfrak{A}_n, \quad |\partial_{t_c}^n b_3| \leq \mathfrak{B}_n, \quad |\partial_{t_c}^n \lambda_3| \leq \mathfrak{D}_n. \end{aligned} \tag{84}$$

The exact, signed amplitude derivative recurrence underlying these bounds is

$$\partial_{t_c}^{n+1} y = \sum_{j=0}^n \binom{n}{j} (\partial_{t_c}^j C_3) (\partial_{t_c}^{n-j} y), \quad n \geq 0. \tag{85}$$

In particular the exact second derivative contains the changing coefficient and the complete diffusion square:

$$\ddot{y} = \begin{pmatrix} \lambda_3^2 + a_3 b_3 - \dot{\lambda}_3 & \dot{a}_3 - 2\lambda_3 a_3 \\ \dot{b}_3 - 2\lambda_3 b_3 & \lambda_3^2 + a_3 b_3 - \dot{\lambda}_3 \end{pmatrix} y. \quad (86)$$

Proof. Introduce the exact auxiliary functions $q = \chi^{-1}$ and $j = R_3^{-1}$, both well-defined by (73). Direct differentiation of their specified inverses gives

$$\dot{q} = -ah\Omega_1 q^3, \quad \dot{j} = -2bg\Omega_1 j^2.$$

These identities retain the actual phase lengths; the symbols do not replace them by constant values. The complete equations for $(h, \Omega_1, A_1, P_2, q, g, N_3, j)$ are therefore the signed polynomial system consisting of (70), (71) and these two equations, with $\chi^2 = h^2 + a^2$ in the parent losses. Their zero-order absolute values are bounded respectively by $(H_0, W_0, A_0^{\text{bd}}, P_0, Q_0, G_0, N_0, J_0)$: the claims for q and j use $\chi \geq \chi_b$ and $R_3 \geq 1$.

For any m differentiable factors the ordinary n th derivative of their product is $\mathcal{P}_n$ applied to their derivative arrays. To prove this formula without assuming it, start with $n = 0$. Differentiating a term of order n once chooses one factor whose index increases. At indices summing to $n + 1$, the resulting coefficient is the sum of $n!/(j_1! \cdots (j_k - 1)! \cdots j_m!)$ over indices k with $j_k > 0$. It is $n!(j_1 + \cdots + j_m)/(j_1! \cdots j_m!) = (n + 1)!/(j_1! \cdots j_m!)$. This proves the formula by induction.

Apply this exact product derivative identity to the signed parent equations and then the triangle inequality. For instance

$$\partial_{t_c}^{n+1} N_3 = -dbK_1^2 \partial_{t_c}^n A_1 - dcK_2^3 \partial_{t_c}^n \{(h^2 + a^2)P_2\},$$

and its absolute value is bounded by the displayed recurrence for N_{n+1} . This example shows explicitly that neither inherited diffusion term is omitted. The other seven equations give exactly (81). Induction in n proves all eight parent derivative bounds. The identities $R_3 = g^2 + b^2$, $a_3 = N_3 j / K_3$, $b_3 = K_3 c q$ and $\lambda_3 = dK_3^2 R_3$ then prove (82) for $n \geq 1$ by the same product formula. Their zero-order bounds were already proved in (73).

Repeatedly differentiate the exact pair to obtain (85). Its componentwise absolute value gives (83), with the zero and first orders supplied by Theorem 1. Induction now proves the two amplitude bounds at every order, and the Euclidean norm bound follows by summing squares. Finally $\ddot{y} = (\dot{C}_3 + C_3^2)y$ and direct multiplication of the two by two matrix give (86). Every recurrence uses finitely many sums at each specified order. Thus each resulting bound is a determinate expression in the original physical constants and actual birth data, rather than an unevaluated supremum of a missing coefficient derivative. $\square$

19.5 Derivative transport through the specified source clock

Let $L_s > 0$ be the source slot length used for this actual interval, and define its specified clock by

$$\alpha_c = \tau_a / L_s, \quad t_c = t_b + \alpha_c v, \quad 0 \leq v \leq L_s.$$

The two actual endpoints are retained. Since α_c is constant for the fixed trajectory and slot, repeated chain differentiation gives, exactly,

$$\partial_v^n y(t_b + \alpha_c v) = \alpha_c^n \partial_{t_c}^n y(t_b + \alpha_c v), \quad |\partial_v^n y|_2 \leq \alpha_c^n \sqrt{T_n^2 + O_n^2}. \quad (87)$$

In a source chart with fixed label Q , exponent h_{src} , and $\partial_t v = Q^{-1-h_{\text{src}}}$, the derivative contribution acting on this amplitude alone is consequently

$$\partial_t^n y(t_b + \alpha_c v) = (\alpha_c Q^{-1-h_{\text{src}}})^n \partial_{t_c}^n y(t_b + \alpha_c v).$$

This identity holds with the stated fixed chart parameters and affine clock; derivatives of any additional frame, cutoff, or varying parameter must be included by the ordinary product rule. In particular neither the clock factor nor the physical frequency has been replaced by one. These finite derivative estimates establish no uniform terminal extension for an infinite sequence.

Part III

Exact clock, signed frame, and actual source pulse

20 The actual third-growth pair in the source clock and frame

This calculation uses equations (6.2)–(6.12), (7.2)–(7.12), and Lemma 7.4 of the retained 166-page source. Its SHA-256 is displayed in two consecutive segments:

```
0e779481c4da40bd28d1e642e1d8ca57
447d129610df28dfa5a11e9af8ae228f.
```

The coupled input is the actual third-growth interval proved in `finite_stage/third_growth_entry.tex`. The source's pulse equation is written at physical viscosity one. We retain an arbitrary positive physical viscosity ν_{NS} when computing a candidate residual; when identifying the source's particular homogeneous pulse, we explicitly set $\nu_{\text{NS}} = 1$ and harmonic $m = 1$. The original coupled time scale $\nu = \sqrt{A_0}$, the physical coupled diffusivity d , and the source chart parameter $\varepsilon = Q^h$ are distinct. The source growth constant denoted λ_0 in $\lambda(v)$ below is the source's own growth constant. It is not identified with the coupled integer frequency $\lambda_0 = \mu = \lceil \sqrt{A_0}/2 \rceil$.

20.1 Exact physical differentiation of the source pulse clock

Fix a source label and one of its local lifts. Set

$$b_g = \sqrt{2} - 1, \quad v_r = (1, -b_g), \quad v_t = (b_g, 1), \quad J_g = \begin{pmatrix} 3 & 1 \\ 1 & 5 \end{pmatrix}, \quad T_g = 4 + \sqrt{2}, \quad \Lambda_g = 4 - \sqrt{2}.$$

Direct multiplication gives $J_g v_t = T_g v_t$ and $J_g v_r = \Lambda_g v_r$, while $v_t \cdot v_r = 0$ and $|v_t|^2 = |v_r|^2 = 1 + b_g^2$. The source retains

$$d_r = 2 \left((1 + h) \frac{\log \Lambda_g}{\log T_g} - h \kappa_s \right), \quad \kappa_s = 10^{-5},$$

$$Q = 2^{-\ell}, \quad i = \left\lfloor \log_{T_g} \frac{Q^{-1-h}}{S_*} \right\rfloor, \quad c_i = T_g^i Q^{1+h}.$$

On the local lift

$$Y_i - c_\gamma - k_{\text{copy}} = \xi_g v_r + \eta_g v_t, \quad v = \frac{\eta_g + r_0}{c_i}, \quad L_s = \frac{2r_0}{c_i}, \quad \eta_g = \frac{v_t \cdot (Y_i - c_\gamma - k_{\text{copy}})}{1 + b_g^2}.$$

The label and integer k_{copy} are held fixed on this lift. Consequently $L_i \eta_g = 0$, $N_i \eta_g = 1$, where $L_i = v_r \cdot \partial_{Y_i}$ and $N_i = v_t \cdot \partial_{Y_i}$. The clock has no explicit dependence on the slow variables R, Z, T . The evaluated chart derivatives are exactly

$$D_r = \partial_R + M_i d_r R^{d_r-1} L_i, \quad D_z = \varepsilon \partial_Z, \quad t_* = -\varepsilon \partial_T + c_i N_i, \quad M_i = \Lambda_g^i Q^{d_r/2}.$$

It follows by applying these displayed operators, without estimating any coefficient, that

$$D_r v = D_z v = 0, \quad t_* v = 1. \tag{88}$$

There is also an explicit physical verification. Evaluate at the source map $Y = v_r r^{d_r} + v_t t$ modulo the lattice. Locally the lift of $Y_i = J_g^i Y$ differs from $\Lambda_g^i v_r r^{d_r} + T_g^i v_t t$ by a constant lattice vector.

The dual formula for η_g therefore gives $\eta_g = T_g^i t + C_\gamma$ with a locally constant C_γ . Thus, on the evaluated local rectangle,

$$\partial_r v = \partial_z v = \partial_\theta v = 0, \quad \partial_t v = \frac{T_g^i}{c_i} = Q^{-1-h}. \quad (89)$$

The ∂_t in this equation is the derivative of the evaluated physical function. On the extended domain the corresponding derivative is $\partial_t + N_{\text{abs}}$. These statements are identical after evaluation; neither drops the auxiliary derivative.

Let t_b and $r_a > 0$ be the actual second return time and the actual third target-crossing elapsed time. Define

$$t_c(v) = t_b + \alpha v, \quad \alpha = \frac{r_a}{L_s} > 0. \quad (90)$$

This parametrizes the entire proved coupled growth interval, with its original endpoints. In physical source variables its derivative is $\partial_t t_c = \alpha Q^{-1-h}$ and all three spatial derivatives vanish. The positive orientation of this clock asserts no sign for a velocity. The coupled amplitudes below are strictly negative.

20.2 The actual frame, its inverse, and the direct candidate

Write the original source phase normal as

$$n = \left(x_0 - v(pF_R + p_z G_R), \frac{p}{R}, p_z - \varepsilon v(pF_Z + p_z G_Z) \right).$$

All p, p_z, x_0, k, Q are the original fixed label constants, including $kp \in \mathbb{Z} \setminus \{0\}$. Set

$$K_a = \frac{n_{\text{tan}}}{|n_{\text{tan}}|}, \quad N_a = ((K_a)_z, -(K_a)_\theta), \quad s_a = \frac{n_r}{|n_{\text{tan}}|},$$

$$c(v) = c_0 \sqrt{1 + s(v)^2}, \quad s(v) = \sigma \left(\frac{u_*}{2} + \frac{u_* v}{L_s} \right).$$

Here $c_0 < 0$ is the unchanged source constant, and K_a, N_a are identified with their tangential three-vectors. In particular $c(v) \neq 0$. Because $p/R \neq 0$, every formula is defined throughout the source rectangle. The actual source frame is

$$U = [e_r - s_a K_a, N_a], \quad M = \begin{pmatrix} 1 & 1 \\ c & -c \end{pmatrix}, \quad B = UM.$$

The two columns of U are orthogonal, with squared lengths $g_a^2 = 1 + s_a^2$ and 1. Both are orthogonal to n . Since $\det M = -2c \neq 0$, B is a bijection from $\mathbb{R}^2$ onto $n^\perp$. The source left inverse, defined on all of $\mathbb{R}^3$, is

$$B_\ell = M^{-1} \begin{pmatrix} e_r^T \\ N_a^T \end{pmatrix}, \quad M^{-1} = \frac{1}{2} \begin{pmatrix} 1 & c^{-1} \\ 1 & -c^{-1} \end{pmatrix}. \quad (91)$$

Indeed the last two rows applied to U give I_2 .

The eigenvalues of

$$B^T B = \begin{pmatrix} g_a^2 + c^2 & g_a^2 - c^2 \\ g_a^2 - c^2 & g_a^2 + c^2 \end{pmatrix}$$

are $2g_a^2$ and $2c^2$, with orthogonal eigenvectors $(1, 1)$ and $(1, -1)$. The Euclidean operator norm and the inverse norm on the tangent plane are therefore exactly

$$\|B\| = \sqrt{2} \max(g_a, |c|), \quad \|(B|_{\mathbb{R}^2})^{-1}\|_{n^\perp \rightarrow \mathbb{R}^2} = \frac{1}{\sqrt{2} \min(g_a, |c|)}. \quad (92)$$

The ambient norm of the particular extension (91) is $(1/\sqrt{2})\max(1, |c|^{-1})$; it need not equal the inverse norm restricted to $n^\perp$.

Retain the actual coupled data

$$K_3 = \mu \hat{\lambda}_3, \quad S_3 = \frac{A_0}{\mu} \hat{\lambda}_3^{-k_3-6}, \quad P_3^* = \frac{A_0}{\mu} \hat{\lambda}_3^{-7/8} = S_3 e^{L_3}, \quad u_i = \sqrt{\frac{b_i}{a_i}} = \frac{K_3}{\sigma_2 \sqrt{n_i}}.$$

On the interval (90), the already constructed solution is

$$y_3(t_c) = \begin{pmatrix} \Theta_3 \\ \Omega_3 \end{pmatrix} = -P(v) \begin{pmatrix} 1 \\ u(v) \end{pmatrix}, \quad S_3 \leq P(v) \leq P_3^*, \quad \frac{u_i}{\sqrt{2}} \leq u(v) \leq 2u_i,$$

with $P(0) = S_3$, $P(L_s) = P_3^*$ and $u(0) = u_i$. These inequalities follow from the actual evolving-parent system, rather than from a new frame approximation. The direct candidate $t_{\text{dir}} = By_3$ consequently has the exact components

$$\begin{aligned} (t_{\text{dir}})_r &= -P(1+u), \\ (t_{\text{dir}})_{\text{tan}} &= Ps_a(1+u)K_a - Pc(1-u)N_a, \\ |t_{\text{dir}}|^2 &= P^2[(1+s_a^2)(1+u)^2 + c^2(1-u)^2], \\ (t_{\text{dir}})_r(t_{\text{dir}})_{\text{tan}} &= -P^2s_a(1+u)^2K_a + P^2c(1-u)^2N_a. \end{aligned} \tag{93}$$

These follow by multiplying $My_3 = -P(1+u, c(1-u))^T$ and using the orthogonality of the two columns of U . In particular

$$|(t_{\text{dir}})_r| \geq S_3(1+u_i/\sqrt{2}), \quad |t_{\text{dir}}| \leq \sqrt{2}\max(g_a, |c|)P_3^* \sqrt{1+4u_i^2}.$$

At the actual target,

$$\frac{P_3^* u_i}{\sqrt{2}} \leq |\Omega_3(t_b + r_a)| \leq 2P_3^* u_i, \quad P_3^* u_i = \frac{A_0}{\sigma_2 \sqrt{n_i}} \hat{\lambda}_3^{1/8}. \tag{94}$$

The last equality retains μ until it cancels against its identical factor in K_3 . Thus a decreasing temperature target does not give a decreasing two-component target.

20.3 The complete principal mismatch, including the cutoff

With a prime denoting v differentiation at fixed slow variables, retain

$$K = \begin{pmatrix} 0 & -2F & 0 \\ 2F + RF_R & 0 & 0 \\ G_R & 0 & 0 \end{pmatrix}, \quad A_\Phi = -K + \frac{n(n^T K - (n')^T)}{|n|^2}, \quad H = B_\ell(A_\Phi B - B').$$

Differentiating $n^T B = 0$ shows $n^T B' = -(n')^T B$. Direct multiplication gives $n^T A_\Phi B = -(n')^T B$ as well. Therefore $A_\Phi B - B'$ takes values in $n^\perp$, and

$$A_\Phi B - B' = BH. \tag{95}$$

The source defines the exact error matrix $E = H - \text{diag}(\lambda, -\lambda)$ with $\lambda = \lambda_0/\sqrt{1+s^2}$. Its source estimate $\|E\| \leq C/S_*$ is not an assertion that E is zero.

The actual coupled matrix and arbitrary-viscosity source loss are

$$C_c = \begin{pmatrix} -d_c & a_3 \\ b_3 & -d_c \end{pmatrix}, \quad d_c = dK_3^2 R_3, \quad a_3 = \frac{N_3}{K_3 R_3}, \quad b_3 = \frac{K_3 c_{\det}}{\chi_{\text{old}}},$$

$$\delta_\nu = \nu_{\text{NS}} m^2 \varepsilon k^2 |n|^2, \quad \Delta_\nu = \delta_\nu - \alpha d_c.$$

Here $c_{\det}$ and χ_{old} are the third-growth determinant constant and older phase length; these names distinguish them from the source $c(v)$ and the cutoff below. Every coupled quantity is evaluated at $t_c(v)$. The actual third growth has zero added amplitude controls, so $y'_3 = \alpha C_c y_3$ exactly.

Let $\chi = \eta(R, Z, T) \chi_g(\xi_g) \psi(v)$ be a smooth scalar cutoff with the source support, and set $t_\chi = \chi B y_3$. The homogeneous-pressure formula, with zero principal source, is

$$\pi_\chi = \frac{i}{km|n|^2} (n \cdot \mathbf{K} t_\chi - n' \cdot t_\chi).$$

The cutoff leaves $n^T t_\chi = 0$. Differentiating this identity and substituting the pressure gives

$$t'_\chi + \mathbf{K} t_\chi + \delta_\nu t_\chi + ikmn \pi_\chi = B \mathcal{E}_\chi, \quad \mathcal{E}_\chi = \chi(\alpha C_c - H + \delta_\nu I) y_3 + \chi' y_3. \quad (96)$$

For completeness, the normal component of the sum before pressure is $-n' \cdot t_\chi + n \cdot \mathbf{K} t_\chi$; multiplication by $ikmn$ times the displayed pressure cancels it. The remaining vector is $t'_\chi - A_\Phi t_\chi + \delta_\nu t_\chi$. Use (95) and $y'_3 = \alpha C_c y_3$ to obtain (96).

In original scalar entries this is

$$\mathcal{E}_\chi = -P \left\{ \chi \begin{pmatrix} \Delta_\nu - \lambda - E_{11} + (\alpha a_3 - E_{12})u \\ \alpha b_3 - E_{21} + (\Delta_\nu + \lambda - E_{22})u \end{pmatrix} + \chi' \begin{pmatrix} 1 \\ u \end{pmatrix} \right\}. \quad (97)$$

All entries are signed. In particular the damping mismatch is $\delta_\nu - \alpha dK_3^2 R_3$, not the sum of the losses. At fixed slow variables and fixed ξ_g , $\chi' = \eta \chi_g \psi'$. The physical derivative of χ also includes $-\varepsilon(\partial_T \eta) \chi_g \psi$ in $t_* \chi$; that slow term belongs to the complete physical residual and has not been included in, or removed by, the principal equation (96). A direct finite bound is

$$|\mathcal{E}_\chi| \leq P \sqrt{1 + u^2} [|\chi| (|\Delta_\nu| + \lambda + \|E\| + \alpha \max(a_3, b_3)) + |\chi'|].$$

This is the triangle inequality and the exact norm $\left\| \begin{smallmatrix} 0 & a_3 \\ b_3 & 0 \end{smallmatrix} \right\| = \max(a_3, b_3)$, with $a_3, b_3 > 0$. It retains the carrier, frame, cutoff and all physical diffusion factors.

20.4 An explicit map of the actual seed to the actual source pulse

For this subsection only, take $m = 1$, $\nu_{\text{NS}} = 1$, so $\delta_1 = \varepsilon k^2 |n|^2$. This is precisely the source homogeneous pulse equation with its given background. Put

$$d_{\text{ref}} = \varepsilon k^2 B_s^2 (1 + s^2), \quad \mathcal{P}_\sigma(v) = \exp \left(\int_{L_s/2}^v (\lambda(w) - d_{\text{ref}}(w)) dw \right).$$

Every coefficient is finite on the closed interval. Thus $\mathcal{P}_\sigma(0) > 0$ exactly, without a Gaussian approximation. Source Lemma 7.4 specifies the initial coordinate vector

$$z_{\sigma,0} = \begin{pmatrix} \mathcal{P}_\sigma(0) \\ 0 \end{pmatrix}, \quad t_\sigma^h(0) = B(0) z_{\sigma,0}.$$

The fixed source label may affect this strictly positive scalar.

For any nonzero real target vector $z_0 = (z_1, z_2)^T$, define

$$L_0 = -\frac{1}{S_3(1+u_i^2)} \begin{pmatrix} z_1 & -z_2 \\ z_2 & z_1 \end{pmatrix} \begin{pmatrix} 1 & u_i \\ -u_i & 1 \end{pmatrix}. \quad (98)$$

This is an explicit invertible linear map with

$$L_0[-S_3(1, u_i)^T] = z_0, \quad \det L_0 = \frac{|z_0|^2}{S_3^2(1+u_i^2)} > 0,$$

and inverse

$$L_0^{-1} = -\frac{S_3}{|z_0|^2} \begin{pmatrix} 1 & -u_i \\ u_i & 1 \end{pmatrix} \begin{pmatrix} z_1 & z_2 \\ -z_2 & z_1 \end{pmatrix}. \quad (99)$$

Indeed the second matrix of (98) sends $(1, u_i)^T$ to $(1+u_i^2, 0)^T$, and each matrix times its transpose is a positive scalar times I_2 . The same multiplication verifies the inverse and proves the exact norms

$$\|L_0\| = \frac{|z_0|}{S_3\sqrt{1+u_i^2}}, \quad \|L_0^{-1}\| = \frac{S_3\sqrt{1+u_i^2}}{|z_0|}. \quad (100)$$

In particular the source target gives

$$L_0 = -\frac{\mathcal{P}_\sigma(0)}{S_3(1+u_i^2)} \begin{pmatrix} 1 & u_i \\ -u_i & 1 \end{pmatrix}, \quad \|L_0\| = \frac{\mathcal{P}_\sigma(0)\mu\hat{\lambda}_3^{k_3+6}}{A_0\sqrt{1+K_3^2/(\sigma_2^2 n_i)}}.$$

No seed or source initial amplitude has been set to one.

Define the two identity-initial-value fundamental matrices by

$$X'_c = \alpha C_c X_c, \quad X_c(0) = I_2, \quad X'_s = (H - \delta_1 I_2) X_s, \quad X_s(0) = I_2.$$

Here is a direct existence, invertibility and uniqueness construction. For a continuous matrix D on $[0, L_s]$ let

$$X_D(v) = I_2 + \sum_{j \geq 1} \int_{0 \leq w_j \leq \dots \leq w_1 \leq v} D(w_1) \cdots D(w_j) dw_j \cdots dw_1.$$

If $\sup \|D\| = M_D$, the j th term is bounded by $(M_D L_s)^j / j!$. Uniform convergence and regrouping the integrals give $X_D = I_2 + \int_0^v D(w) X_D(w) dw$, hence the differential equation. The corresponding right-ordered series for $Z' = -ZD$, $Z(0) = I_2$, converges by the same bound. Differentiation gives $(ZX_D)' = 0$, hence $ZX_D = I_2$. For square matrices this is invertibility. If two solutions agree initially, the difference satisfies the zero-initial-value integral equation; iteration bounds its norm by a fixed finite bound times $(M_D v)^j / j!$ for every j , which tends to zero. This proves uniqueness. Smooth coefficient dependence follows from the same series with the following explicit bound. On a compact coefficient-parameter neighborhood let $M_N \geq 1$ bound every coefficient derivative through order N . Differentiating a product of j factors N times allocates each derivative to one of j factors, so there are at most j^N terms, each of norm at most M_N^j . The differentiated j th integral is therefore bounded by $j^N (M_N L_s)^j / j!$, a summable sequence. Uniform convergence of these derivative series justifies termwise differentiation: integrate each first-derivative series along a parameter segment and interchange the uniform sum with that integral, then iterate through order N . The right-ordered inverse series has the same bound. Pulse derivatives then follow from

the differential equations. This proves smoothness, with the explicit derivative equations below providing further bounds.

Now set

$$L(v) = X_s(v)L_0X_c(v)^{-1}, \quad L^{-1} = X_cL_0^{-1}X_s^{-1}. \quad (101)$$

Differentiation gives

$$L' = (H - \delta_1 I_2)L - \alpha LC_c.$$

The original solution is $y_3(v) = X_c(v)y_3(0)$. Consequently $L(v)y_3(v) = X_s(v)z_{\sigma,0}$, and

$$B(v)L(v)y_3(v) = t_\sigma^h(v) \quad (102)$$

by uniqueness of the source homogeneous equation and the exact initial seed calculation. Thus this finite map reaches the actual source pulse. It does not merely send the coupled pair into an unspecified transverse solution. Multiplying both sides by the identical source cutoffs preserves equality of the amplitudes, and applying identical linear curl and pressure constructions preserves equality of the resulting fields. Multiplication by the source prescribed scalar leading-wave amplitude also preserves that equality. The cutoff principal residual is then $\chi'BLy_3$; the remaining source construction uses its own estimates for that exact source object.

20.5 All scale costs of the finite intertwiner

Here and below $\|\cdot\|$ is the Euclidean matrix norm. Put $D_i = \text{diag}(1, u_i)$, $\kappa_i = \max(u_i, u_i^{-1})$, and

$$\beta_c(v) = \frac{1}{2} \left(a_3(t_c(v))u_i + \frac{b_3(t_c(v))}{u_i} \right).$$

The actual coefficient bounds $a_3 \leq a_i$, $b_3 \leq b_i$ and $a_i u_i = b_i / u_i = \gamma_i$ give $0 < \beta_c \leq \gamma_i$. Conjugating the coupled matrix by the constant D_i gives

$$D_i^{-1}C_c D_i = \begin{pmatrix} -d_c & a_3 u_i \\ b_3 / u_i & -d_c \end{pmatrix}.$$

Its symmetric part has eigenvalues $-d_c \pm \beta_c$. For any solution w in these coordinates, differentiating $|w|^2$ therefore gives

$$2\alpha(-d_c - \beta_c)|w|^2 \leq (|w|^2)' \leq 2\alpha(-d_c + \beta_c)|w|^2.$$

Multiplication by the corresponding scalar integrating factors proves the following bounds, including the lower singular-value estimate needed for the inverse:

$$\|X_c(v)\| \leq \kappa_i e^{\int_0^v \alpha(\beta_c - d_c) dw}, \quad \|X_c(v)^{-1}\| \leq \kappa_i e^{\int_0^v \alpha(\beta_c + d_c) dw}.$$

For $D_s = H - \delta_1 I_2 = \text{diag}(\lambda, -\lambda) - \delta_1 I_2 + E$, its symmetric part is bounded between $(-\lambda - \delta_1 - \|E\|)I_2$ and $(\lambda - \delta_1 + \|E\|)I_2$. The identical energy argument gives

$$\|X_s(v)\| \leq e^{\int_0^v (\lambda - \delta_1 + \|E\|) dw}, \quad \|X_s(v)^{-1}\| \leq e^{\int_0^v (\lambda + \delta_1 + \|E\|) dw}.$$

Combining these four proved estimates with (101) yields

$$\begin{aligned} \|L(v)\| &\leq \kappa_i \|L_0\| e^{\int_0^v [\lambda - \delta_1 + \|E\| + \alpha(\beta_c + d_c)] dw}, \\ \|L(v)^{-1}\| &\leq \kappa_i \|L_0^{-1}\| e^{\int_0^v [\lambda + \delta_1 + \|E\| + \alpha(\beta_c - d_c)] dw}. \end{aligned} \quad (103)$$

All quantities on the right are the retained, finite coefficient paths. In particular the exact coupled diffusion integral at the endpoint is

$$\int_0^{L_s} \alpha d_c dv = dK_3^2 \int_0^{r_a} R_3(t_b + r) dr \in [dK_3^2 r_a, 2dK_3^2 r_a].$$

The equality is the substitution $r = \alpha v$; the interval is the proved actual phase-length bound. The determinant is also exact:

$$\det L(v) = \frac{|z_{\sigma,0}|^2}{S_3^2(1 + u_i^2)} \exp \left(\int_0^v [\operatorname{tr} E - 2\delta_1 + 2\alpha d_c] dw \right). \quad (104)$$

For a differentiable 2 by 2 matrix, expanding its two-column determinant and inserting $X' = DX$ gives $(\det X)' = (\operatorname{tr} D) \det X$. Its initial determinant is one. Apply this calculation to X_s and X_c , and use $\operatorname{tr} H = \operatorname{tr} E$ and $\operatorname{tr} C_c = -2d_c$ to prove (104).

Higher pulse derivatives have the explicit recurrence

$$L^{(n+1)} = \sum_{j=0}^n \binom{n}{j} \left[D_s^{(j)} L^{(n-j)} - L^{(n-j)} D_c^{(j)} \right], \quad D_c = \alpha C_c.$$

For a coefficient parameter derivative ∂_ξ the exact inhomogeneous equation is

$$(\partial_\xi L)' = D_s \partial_\xi L - (\partial_\xi L) D_c + (\partial_\xi D_s) L - L(\partial_\xi D_c).$$

Its initial value is the derivative of the explicit (98). To see directly that this produces a finite bound, multiply on the left by X_s^{-1} and on the right by X_c , differentiate, and integrate the last two terms with these same multipliers. The preceding norm estimates bound every resulting integral on $[0, L_s]$. Iteration proves the analogous assertion for every finite mixed derivative; all product-rule terms and initial derivatives are retained. This also proves smoothness on each compact source slow neighborhood where the frame is defined.

The factor $\mathcal{P}_\sigma(0)/S_3$ and the two explicit propagation costs in (103) can be very large or very small. There is no supplied relation identifying L_s with L_3 . Thus these equations prove a finite invertible map and exact matching to the source object, while exposing the cost of that map. They do not prove a uniform weighted-class estimate for L or for the direct candidate $L = I$. The source's particular pulse retains its own source estimates because of the equality (102). Transport of the full viscosity-one witness to another physical viscosity uses the already proved affine orthogonal and viscosity map. Alternatively one may replace δ_1 by δ_ν in this finite ODE construction, but the resulting changed-background homogeneous pulse is not asserted to be the source's published witness.

Part IV

Complete physical curl, moments, and force costs

21 Actual third-growth budgets in the physical curl and force

We now use the actual third-growth pair and the original source frame of the preceding calculations, with the direct coefficient choice $L = I_2$. This fixes a particular candidate within the earlier general bridge. All source band, box, sign and nonzero harmonic indices below are retained, and only a finite collection is used on the present compact interior patch. The cutoff of each term has compact support inside its source rectangle and active annulus, with smooth extension by zero. The original coupled datum, its selected return time t_b and its growth duration r_a are fixed before assigning these source labels; they have no dependence on the source spatial or auxiliary variables. The source time and the coupled time have the exact dictionary proved above:

$$t_{c,j} = t_b + \alpha_j v_j, \quad \alpha_j = \frac{r_a}{L_{s,j}}, \quad \nabla_x t_{c,j} = 0, \quad \partial_t t_{c,j} = \alpha_j Q_j^{-1-h}.$$

These derivative identities concern the complete physical phase evaluation on each lifted rectangle. They remain valid when differentiating a field supported strictly inside that rectangle: the field vanishes on a collar where its extension would otherwise be needed.

Write $P(t_c) = -\Theta_3(t_c)$, $u(t_c) = \Omega_3(t_c)/\Theta_3(t_c)$. The actual finite third growth gives

$$S_3 \leq P \leq P_3^*, \quad \frac{u_i}{\sqrt{2}} \leq u \leq 2u_i, \quad u_i = \sqrt{b_i/a_i} = \frac{K_3}{\sigma_2 \sqrt{n_i}}, \quad \gamma_i = \sqrt{a_i b_i}.$$

The original quantities remain

$$K_3 = \mu \hat{\lambda}_3, \quad S_3 = \frac{A_0}{\mu} \hat{\lambda}_3^{-k_3-6}, \quad P_3^* = \frac{A_0}{\mu} \hat{\lambda}_3^{-7/8}, \quad \nu = \sqrt{A_0}.$$

In particular the following two component budgets have different original frequency powers:

$$\begin{aligned} Y_1 &= P_3^* = \frac{A_0}{\mu} \hat{\lambda}_3^{-7/8}, \\ Y_2 &= 2u_i P_3^* = \frac{2A_0}{\sigma_2 \sqrt{n_i}} \hat{\lambda}_3^{1/8}, \\ V_1 &= 2\gamma_i P_3^*, \quad V_2 = \gamma_i u_i P_3^* = \frac{\gamma_i A_0}{\sigma_2 \sqrt{n_i}} \hat{\lambda}_3^{1/8}. \end{aligned} \tag{105}$$

Here the indices 1, 2 mean the two amplitude components, not physical vector directions or radial moment weights. The derivative calculation above proves $|y_a(t_c)| \leq Y_a$ and $|\dot{y}_a(t_c)| \leq V_a$ on the entire original interval, including its original negative seed. For the second component at the actual threshold, the sharper two-sided conclusion is

$$\frac{A_0}{\sigma_2 \sqrt{2n_i}} \hat{\lambda}_3^{1/8} \leq |\Omega_3(t_c^a)| \leq \frac{2A_0}{\sigma_2 \sqrt{n_i}} \hat{\lambda}_3^{1/8}.$$

All these identities follow by multiplying the original quotient bounds by P and using $K_3 = \mu \hat{\lambda}_3$; the factor μ is canceled only in the explicitly displayed product $u_i P_3^*$, not removed from the underlying objects or their definitions.

21.1 Two complete spatial shape fields for each retained harmonic

For harmonic j , retain $k_j m_j \neq 0$, where k_j is the source carrier and m_j its integer harmonic. Neither equals the coupled integer k_3 by definition. Let B_j be the source frame, n_j its normal, and χ_j the full real slow, transverse and pulse cutoff. All are the original specified functions on that label. Put, for $a = 1, 2$ and the standard vectors $e_a \in \mathbb{R}^2$,

$$\begin{aligned} T_{ja} &= \chi_j B_j e_a, & C_{ja} &= \frac{i n_j \times T_{ja}}{k_j m_j |n_j|^2}, \\ W_{ja} &= \text{Re}[(T_{ja} + \text{curl}_{*,j,\text{coeff}} C_{ja}) e^{ik_j m_j \Phi_j}], \\ \varpi_{ja} &= \text{Re} \left[\frac{i \{n_j^T K_j T_{ja} - (n'_j)^T T_{ja}\}}{k_j m_j |n_j|^2} e^{ik_j m_j \Phi_j} \right], \\ Z_{ja} &= Q_j^{-A} W_{ja}, & \Pi_{ja} &= Q_j^{-2A} \varpi_{ja}, & A &= \frac{1}{2} + h. \end{aligned} \tag{106}$$

Vectors in a cross product have their cylindrical components in the oriented orthonormal physical basis. The coefficient curl is the full operator from the preceding moment calculation. Equivalently,

$$Z_{ja} = \text{Re} \text{curl}_x [Q_j^{1/2-A} C_{ja} e^{ik_j m_j \Phi_j}].$$

The transverse triple product proves this equivalence, including every cutoff derivative. Thus each Z_{ja} is a smooth compactly supported divergence-free physical vector field; the same divergence identity holds before auxiliary evaluation, with the commuting lifted derivatives. The pressure coefficient is linear in T_{ja} and includes the full moving-normal term.

The actual added physical fields, with the two original amplitudes, are exactly

$$w_y = \sum_j \sum_{a=1}^2 y_a(t_{c,j}) Z_{ja}, \quad \pi_y = \sum_j \sum_{a=1}^2 y_a(t_{c,j}) \Pi_{ja}. \tag{107}$$

Indeed $\nabla_x t_{c,j} = 0$ gives $\text{curl}(y_a(t_{c,j}) \mathcal{A}) = y_a(t_{c,j}) \text{curl} \mathcal{A}$. It also gives $\nabla(y_a(t_{c,j}) \Pi) = y_a(t_{c,j}) \nabla \Pi$. The pressure formula in (106) is linear, so it agrees with the selected pressure for the amplitude $\chi_j B_j y(t_{c,j})$. This proves (107); in particular it does not discard a spatial derivative of a rapidly growing amplitude. That derivative has been calculated and is zero in these precise source coordinates.

The exact spatial derivative is

$$\partial_z w_y = \sum_{j,a} Q_j^{-A-D} y_a(t_{c,j}) \partial_{Z_j} W_{ja}, \quad D = \frac{1}{2} - h,$$

where $\partial_{Z_j} W_{ja}$ differentiates its coefficient, its exponential phase and all source frame and cutoff factors. The absolute auxiliary evaluation map is independent of z . There is no derivative of the actual coupled time in this formula.

21.2 Quantitative radial moments with the original component powers

For a physical field f , use the precise norm

$$\|f\|_{[e]}^2 = \int_0^\infty r^e \langle |f|^2 \rangle dr, \quad e = 1, 2,$$

with the normalized angular and auxiliary averages of the moment calculation. In a fixed band let $\|F\|_{[e],j}^2 = \int R_j^e \langle |F|^2 \rangle dR_j$. All norms are at the corresponding fixed slow axial coordinate and physical time. Define the actual finite shape budgets

$$\begin{aligned}\mathcal{V}_e &= \sum_{j,a} Y_a Q_j^{(e+1)/4-A} \|W_{ja}\|_{[e],j}, \\ \mathcal{Z}_e &= \sum_{j,a} Y_a Q_j^{(e+1)/4-A-D} \|\partial_{Z_j} W_{ja}\|_{[e],j}.\end{aligned}\tag{108}$$

The factors Y_1, Y_2 are the two explicit original-frequency expressions in (105), not a common unspecified constant. Triangle inequality, the pointwise amplitude bounds, and $r^e dr = Q_j^{(e+1)/2} R_j^e dR_j$ prove

$$\|w_y\|_{[e]} \leq \mathcal{V}_e, \quad \|\partial_z w_y\|_{[e]} \leq \mathcal{Z}_e.$$

No orthogonality of distinct harmonics has been assumed in these bounds. One may evaluate every norm by the explicit source fields (106); compactness makes each finite.

Let $w_* = u_* - \langle u_* \rangle_\theta$ be the actual old wave, including its curl corrections, and set

$$\mathcal{W}_e = \|w_*\|_{[e]}, \quad \mathcal{W}_{e,z} = \|\partial_z w_*\|_{[e]},$$

restricting the old-field radial integrals to a fixed enclosing support of the new fields and their derivatives. The exact moments already proved are

$$\begin{aligned}M_2 &= \partial_z \int r^2 \langle w_{*,z} w_{y,\theta} + w_{y,z} w_{*,\theta} + w_{y,z} w_{y,\theta} \rangle dr, \\ M_1 &= \partial_z \int r \langle 2w_{*,z} w_{y,z} + w_{y,z}^2 \rangle dr.\end{aligned}$$

Apply the product rule to every term and then Cauchy–Schwarz on its original weighted product measure. This gives the concrete simultaneous bounds

$$|M_e| \leq 2(\mathcal{W}_{e,z} \mathcal{V}_e + \mathcal{W}_e \mathcal{Z}_e + \mathcal{V}_e \mathcal{Z}_e), \quad e = 1, 2.\tag{109}$$

For M_2 , the first two cross products contribute one term each to each of the first two summands, and the self product contributes two terms to the last. For M_1 the explicit coefficient 2 and the derivative of the square give exactly those same three coefficients. Thus (109) retains the complete cross-and-quadratic dependence on the original component budgets.

21.3 The full physical residual with all clock and diffusion costs

Fix the actual finite source state (u_*, p_*) used in the bridge and an arbitrary physical viscosity $\nu_{\text{NS}} > 0$ in the residual. It remains distinct from the coupled d and $\nu = \sqrt{A_0}$. Define the explicitly differentiated shape residual

$$\mathcal{K}_{ja} = \partial_t Z_{ja} + (u_* \cdot \nabla) Z_{ja} + (Z_{ja} \cdot \nabla) u_* - \nu_{\text{NS}} \Delta Z_{ja} + \nabla \Pi_{ja}.$$

With $C_3(t_c) = \begin{pmatrix} -dK_3^2 R_3 & a_3 \\ b_3 & -dK_3^2 R_3 \end{pmatrix}$, the full residual increment of (107) is

$$\begin{aligned}\Delta R_y &= \sum_{j,a} y_a(t_{c,j}) \mathcal{K}_{ja} \\ &\quad + \sum_{j,a} \alpha_j Q_j^{-1-h} (C_3 y)_a(t_{c,j}) Z_{ja} \\ &\quad + \sum_{j,a,k,b} y_a(t_{c,j}) y_b(t_{c,k}) (Z_{ja} \cdot \nabla) Z_{kb}.\end{aligned}\tag{110}$$

To prove it, substitute (107) into the complete Navier–Stokes residual and subtract $R_{\nu_{\text{NS}}}(u_*, p_*)$. The time product rule gives $\partial_t y_a = \alpha_j Q_j^{-1-h} (C_{3y})_a$. Every spatial derivative of $y_a(t_{c,j})$ vanishes, so neither advection nor diffusion nor pressure adds an unrecorded derivative of that factor. The three linear spatial terms, the time derivative of Z_{ja} and the pressure gradient are precisely $\mathcal{K}_{ja}$. The remaining quadratic product is the last full sum in (110). This proves the identity with every old-wave interaction, cutoff, curl term and pressure included. Distinct labels with disjoint source supports have zero corresponding products; no such cancellation is used for harmonics of the same label. The baseline $R_{\nu_{\text{NS}}}(u_*, p_*)$ must be added back to recover the candidate's total residual.

The source units make every factor in this identity explicit. Let $\nabla_{*,j}$ and $\Delta_{*,j}$ be the full cylindrical spatial operators, including basis derivatives and the lifted auxiliary derivatives, with $\nabla_x = Q_j^{-1/2} \nabla_{*,j}$ and $\Delta_x = Q_j^{-1} \Delta_{*,j}$ on that chart. Put $U_{*,j} = Q_j^A u_*$ and $\mathbf{t}_{*,j} = Q_j^{1+h} \partial_t$ with the full physical chain rule. Its extended expression is $-\varepsilon_j \partial_{T_j} + c_{i,j} N_{i,j}$ as in source (6.6). Define

$$\mathfrak{K}_{ja} = \mathbf{t}_{*,j} W_{ja} + (U_{*,j} \cdot \nabla_{*,j}) W_{ja} + (W_{ja} \cdot \nabla_{*,j}) U_{*,j} - \nu_{\text{NS}} \varepsilon_j \Delta_{*,j} W_{ja} + \nabla_{*,j} \varpi_{ja}.$$

Each term is obtained by differentiating the explicit shapes in (106), and

$$\mathcal{K}_{ja} = Q_j^{-2A-1/2} \mathfrak{K}_{ja}.$$

For the time term the exponent identity is $A + 1 + h = 2A + 1/2$; for viscosity it is $Q_j^{-A-1} = Q_j^{-2A-1/2} \varepsilon_j$. The nonlinear and pressure terms have the same stated residual unit. This proves every factor rather than replacing the source or coupled viscosity by a common normalized number.

Using Euclidean vector norms and the induced norm of the differentiated vector map, (110) and (105) give the following pointwise bound in actual shape fields:

$$\begin{aligned} |\Delta R_y| &\leq \sum_{j,a} Q_j^{-2A-1/2} (Y_a |\mathfrak{K}_{ja}| + \alpha_j V_a |W_{ja}|) \\ &\quad + \sum_{j,a,k,b} Y_a Y_b Q_j^{-A} Q_k^{-A-1/2} |W_{ja}| \|\nabla_{*,k} W_{kb}\|. \end{aligned} \tag{111}$$

Both Q_j and Q_k remain in every mixed-band term. The expression $\nabla_{*,k} W_{kb}$ includes the cylindrical frame connection, so the bound controls the actual Cartesian directional derivative. The clock satisfies the explicit original-stage bounds

$$\frac{L_3}{2\gamma_i L_{s,j}} \leq \alpha_j \leq \frac{8\sqrt{2}L_3}{\gamma_i L_{s,j}}.$$

Consequently its derivative costs in the first line can, if desired, be bounded using the original constants by

$$\alpha_j V_1 \leq \frac{16\sqrt{2}L_3}{L_{s,j}} \frac{A_0}{\mu} \hat{\lambda}_3^{-7/8}, \quad \alpha_j V_2 \leq \frac{8\sqrt{2}L_3}{L_{s,j}} \frac{A_0}{\sigma_2 \sqrt{n_i}} \hat{\lambda}_3^{1/8}.$$

These are finite quantitative costs for the actual inserted third stage. They do not assert decay of the resulting residual at a singular limit.

The source pulse equation (7.5) itself belongs to its viscosity-one source object. At a general ν_{NS} in the displayed candidate residual, the principal wave damping is $\nu_{\text{NS}} m_j^2 \varepsilon_j k_j^2 |n_j|^2$. Alternatively, the exact viscosity transport already proved can be applied to the complete viscosity-one source field and this complete candidate. That transport rescales support, force, residual and vorticity together. Merely changing this one damping coefficient does not assert that the unchanged source background is a solution with the same force at the changed viscosity.

Equations (106), (107), (109) and (110) are the explicit substitution of the actual third-growth amplitudes into the source-coordinate construction. Every surviving moment and residual has an actual differentiated field and a retained-scale bound. The exact intertwiner above supplies the relation to the source's homogeneous pulse; neither this direct candidate nor that finite coordinate identity is silently promoted to a completed infinite modified coupled construction.